Managerial Coordination in Decentralized Markets

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March 2026

Abstract

This paper develops a model of managerial coordination in decentralized relational contracting markets. Managers aggregate fluctuating buyer demand and coordinate the assignment of producers, lengthening productive relationships and relaxing dynamic enforcement constraints. This demand-smoothing benefit must be weighed against double marginalization: managers themselves require incentive rents. The trade-off endogenously sorts buyers into managerial coordination or decentralized exchange. In equilibrium, managed producers earn lower, more compressed pay with lower separation rates. The introduction of managers increases trade but redistributes surplus from producers to buyers. We extend the model to show that managerial coordination promotes specialization. The framework applies to professional service firms, global sourcing intermediaries, and digital matching platforms.

Keywords: managerial coordination, relational contracts
JEL: D23, L22, L24

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1 Introduction

In many industries, production is coordinated not through direct bilateral exchange but through centralized intermediaries who monitor performance and allocate assignments across large numbers of workers or suppliers (Chandler 1977; Alchian and Demsetz 1972). The prevalence of such managerial coordination is striking because, in principle, buyers and producers could transact directly. Why do parties voluntarily cede coordination to a third party, paying for an additional layer of intermediation?

Standard theories offer partial answers rooted in transaction costs, property rights, and hold-up problems (e.g., Coase 1937; Williamson 1975; Grossman and Hart 1986). But these frameworks typically abstract from the repeated nature of business relationships and from the matching markets in which they form. A complementary perspective, grounded in relational contracting (e.g., Levin 2003; Malcomson 2013; Gil and Zanarone 2017), emphasizes that when formal contracts are incomplete, parties sustain cooperation through the promise of future surplus. This literature has made significant progress in bilateral and market settings (e.g., MacLeod and Malcomson 1989; Board and Meyer-ter-Vehn 2015; Fahn and Murooka 2025), yet a basic question remains unanswered: if agents can contract with one another directly, why does centralized coordination add value?

This paper develops a tractable repeated-game model of managerial coordination in decentralized matching markets. The core of our analysis is a novel economic mechanism: in settings characterized by moral hazard and demand fluctuations, managers can reduce the cost of motivating performance by coordinating the assignment of buyers to producers. We characterize the equilibrium effects of this form of coordination: managed producers earn lower, more compressed pay with lower separation rates, while the introduction of managers increases trade but redistributes surplus from producers to buyers. Our work extends the literature on intermediaries (e.g., Rubinstein and Wolinsky 1987; Biglaiser 1993; Spulber 1999) by isolating a distinct channel: managers add value by pooling demand and thereby relaxing relational incentive constraints, even in the absence of search frictions or adverse selection.

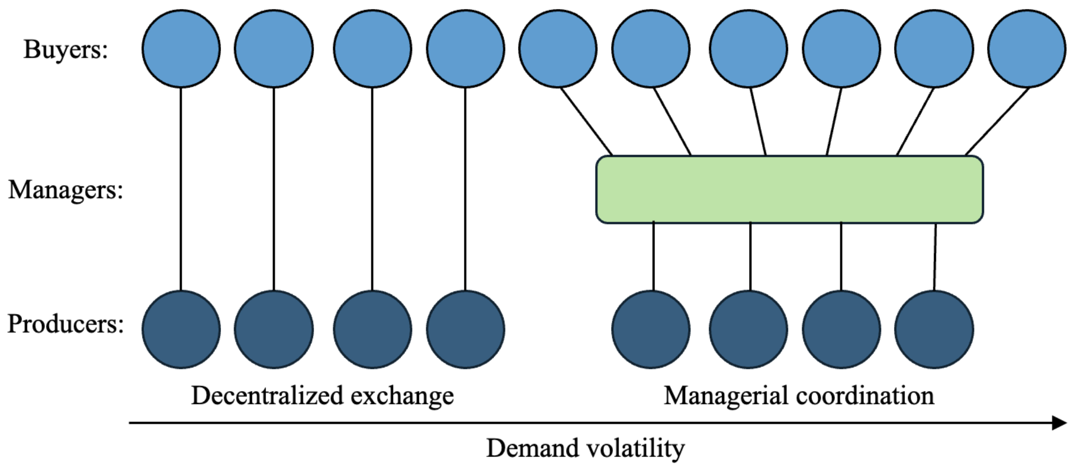
Our framework builds on MacLeod and Malcomson (1989), who study relational contracts in decentralized matching markets. As in their model, formal contracts cannot enforce effort, so buyers motivate producers through the promise of future surplus. Matching is frictionless, as in Board and Meyer-ter-Vehn (2015): there are no search delays or hidden payoff-relevant information. We depart from prior models by introducing fluctuating buyer demand, a realistic feature of repeated business interactions (Macchiavello and Morjaria 2015), and managers who cannot themselves produce but can monitor and allocate a large number of producers. In equilibrium, managers enter relational contracts on both sides of the market, aggregate fluctuating demand, coordinate assignments, and keep track of producer performance.

The model reveals that managerial coordination involves a trade-off between demand smoothing and double marginalization. The benefit is that by aggregating demand and coordinating assignments, managers prevent producers from becoming idle or unmatched. Longer productive horizons relax the dynamic enforcement constraints—the incentive conditions requiring that producers prefer exerting effort over shirking—and reduce the incentive rent needed to motivate producers to exert effort. However, managers must themselves be motivated by an additional incentive rent. The cost of managers is thus a form of double marginalization.

Because of this trade-off, buyers sort into managerial coordination and decentralized exchange. Buyers with short-lived demand contract with managers, while buyers with long-lived demand contract directly with producers. The sorting depends on both individual demand transience and market tightness: if the market is sufficiently tight, managers emerge as centralized contracting nexuses. Figure 1 illustrates the relational contracting networks that emerge in the unique steady-state equilibrium.

We derive comparative statics on the determinants and impacts of this form of managerial coordination. First, directly contracted producers have higher average pay, more dispersed pay, higher separation rates, and higher idleness than managed producers. Second, the introduction of managers into an economy benefits buyers with fluctuating demand,

Figure 1: Endogenous relational contracting networks in steady-state equilibrium



but reduces the pay of producers initially earning pay premiums. As such, managerial coordination both increases and redistributes total surplus.

In an extension, we show that managerial coordination promotes specialization: it increases the measure of specialist producers, and buyers with larger gains from specialization are more likely to choose managerial coordination. In another extension, we consider a version of the model where the producer population is exogenous, in contrast to our baseline model, where it is determined through the free entry of producers.

We illustrate how the framework can shed light on modern organizational forms by discussing three empirical applications: (i) professional service firms that monitor and deploy large numbers of workers (e.g., legal, accounting, IT, and HR firms), (ii) global sourcing firms (e.g., Li & Fung and Shein) that manage vast supplier networks and coordinate significant shares of world production, and (iii) digital matching platforms like Uber and Airbnb, which allocate and monitor (i.e., manage) growing numbers of independent contractors.

Throughout the paper, we connect the model's predictions to the empirical literature on professional service firms. We focus on professional service firms because the increasing availability of employee-employer matched data has made micro-evidence on the topic particularly abundant. As we show, the model generates predictions consistent with recent empirical findings on the drivers and effects of professional service outsourcing.

Related literature. Our paper contributes to the literature on relational contracts, and in particular to the sub-literature on multilateral relational contracts in fluctuating environments.[1] Board (2011) shows that the tension between incentive provision and flexible allocations can lead buyers to restrict the number of trading partners. Andrews and Barron (2016) show that it can cause dynamic allocations to depend on payoff-irrelevant past performance. Li and Powell (2020) highlight that a similar tension can lead agents to interact in multiple activities. We show that the same fundamental tension can induce agents in matching markets to delegate allocation responsibilities to a centralized coordinator. To show this, we build on the game-theoretic machinery developed in the literature on relational contracts in matching markets (Shapiro and Stiglitz 1984; MacLeod and Malcomson 1989, 1998; Yang 2008; Board and Meyer-ter-Vehn 2015; Fahn 2017; Powell 2019; Li 2022; Fahn and Murooka 2025). To our knowledge, we are the first to show how centralized managerial coordination reduces incentive costs in fluctuating matching markets for relational contracts and thereby increases trade and specialization.

Our model can be viewed as providing a novel, parsimonious, micro-founded answer to a variant of the question first posed by Coase (1937): Why do centralized resource allocators emerge in decentralized markets? Our answer is closely related to informal intuitions in early contract-based theories of the firm. Alchian and Demsetz (1972) emphasized that the presence of a central contracting party who monitors performance and can exclude underperforming team members is the essence of the firm. Jensen and Meckling (1976, pp. 310–311) similarly argued that "contractual relations are the essence of the firm, not only with employees but with suppliers, customers, creditors, and so on," and Demsetz (1988, p. 154) described the firm as a "nexus of contracts" whose key attributes are continuity of association, specialization, and managerial direction. Our model formalizes these intuitions in a dynamic setting with imperfect enforcement, clarifying when and why centralized

[1] For surveys of recent work on relational contracts in economics, see MacLeod (2007), Malcomson (2013), Gil and Zanarone (2017), Macchiavello and Morjaria (2023), and Muehlheusser et al. (2023).

contracting intermediaries arise to coordinate allocations.

Our work extends the literature on intermediaries, which to date has focused largely on middlemen who alleviate either search frictions (e.g. Rubinstein and Wolinsky 1987) or adverse selection (e.g. Biglaiser 1993).[2] We formalize a different mechanism: intermediaries may overcome moral hazard by coordinating buyer–producer assignments in a relational contracting environment. Crucially, this mechanism operates even in the absence of search, bargaining, and contract-writing costs, as well as hidden information about agent types. In a closely related work, Belavina and Girotra (2012) develop a model of relational intermediaries in supply chains with two buyers, two sellers, and non-transferable utilities. Our transferable-utility framework allows us to characterize the determinants and welfare consequences of this form of intermediation in a general equilibrium setting.

Our work can be viewed as a novel, micro-founded general equilibrium model of managerial talent. In Lucas (1978), managerial talent is scarce and complementary to production workers, generating high managerial pay. Garicano (2000) models knowledge hierarchies in which managers act as expert problem-solvers for less skilled workers (see also Garicano and Rossi-Hansberg 2015). These approaches emphasize technological complementarities between managers and workers but abstract from incentive problems. Our model complements this literature by grounding managerial coordination in relational incentive constraints, while similarly yielding predictions about organizational structure and distributional outcomes in market equilibrium.

Finally, our work extends the literature on market institutions that support cooperation. Milgrom et al. (1990) study private judges in medieval trade in a repeated Prisoner's Dilemma, showing that such institutions can create trust with less observability than decentralized reputation systems (see also Greif 1993, 2006; Greif et al. 1994; Calvert 1995; Fafchamps 2004). We follow a similar repeated-game logic but shift the focus from histori-

[2]On search-based intermediation, see, for example, Rubinstein and Wolinsky (1987), Gehrig (1993), Yavaş (1994, 1996), Rust and Hall (2003), Antras and Costinot (2011), Wright and Wong (2014), Nosal et al. (2015), Nosal et al. (2019), and Rhodes et al. (2021). On adverse selection and information provision by intermediaries, see, for example, Biglaiser (1993), Lizzeri (1999), Glode and Opp (2016), and Biglaiser and Li (2018).

cal or legal institutions to managerial coordination, arguing that the need for trust remains pervasive even with modern courts and regulation.

Roadmap. Section 2 introduces our model and empirical applications. Section 3 characterizes and compares direct and managed contracts, taking matching market conditions as given. Section 4 derives the steady-state market equilibrium. Section 5 pursues extensions. Section 6 concludes.

2 Model

Basics. Time is discrete and infinite, $\tau \in \{0, 1, \ldots\}$. Buyers are heterogeneous, indexed by i, and have unit mass. Producers are identical and have measure n. Managers are identical and have a finite number K. All players are infinitely-lived. They have a common discount factor $\delta \in (0,1)$. The measure of producers, n, is determined by endogenous entry subject to entry cost $C > 0$. Entry cost C represents training or opportunity cost.

Demand realization. At the start of each period, the demand of each buyer i, denoted as $d_{i,\tau} \in \{0,1\}$, is realized and publicly observed.[3] Each buyer's demand $d_{i,\tau}$ is independently drawn following a Markov process. The demand-switching probabilities are determined by each buyer's publicly known type $\alpha_i = (\alpha_{1i}, \alpha_{0i})$: with probability α_{1i}, buyer i's demand switches from 1 to 0; with probability α_{0i}, buyer i's demand switches back from 0 to 1. The buyers types have distribution F on $[0,1] \times [0,1]$.

Matching. After demand is realized, matched players may choose to separate. Buyers may then open vacancies for either producers or managers. Since managers cannot produce by themselves, they fulfill buyer demand by matching with producers in turn.

[3] To focus on the relational incentives, we abstract from any adverse selection problem in the model. Therefore, when contracting with buyers, producers know whom they are dealing with and have the correct expectation of how the relationships will go.

The timing is as follows: (1) buyers open either manager vacancies or producer vacancies, (2) managers are matched with manager vacancies, (3) managers open producer vacancies, and (4) unmatched producers are matched with producer vacancies. Each buyer may open at most one vacancy, while managers may match with a continuum of buyers and producers.

Following Board and Meyer-ter-Vehn (2015), the matching of producer vacancies follows three axioms. First, matching is *frictionless*, meaning that all vacancies are filled subject to the managers' or the producers' participation constraints. Second, matching is *anonymous*, meaning that each producer's probability of becoming matched with a vacancy does not depend on their past behavior. The assumption of anonymous matching for producers is standard and simplifies analysis.[4] Third, matching is *random*, i.e., unmatched producers have the same probability of becoming matched.[5]

The matching of manager vacancies is assumed to be frictionless, random, but *not* anonymous. In other words, if a manager violates its relational contract with a buyer and becomes unmatched, the buyer remembers the manager's violation and refuses to match with the manager forever. This assumption captures the idea that the market for managers is comparatively small.

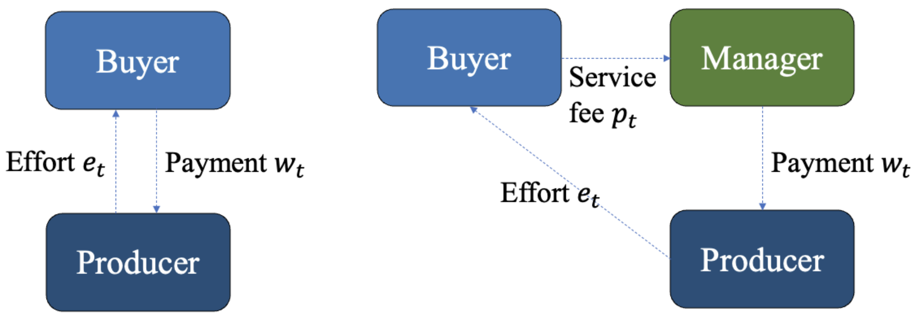
We call a buyer-producer match a *direct match*, since it does not involve a manager. A buyer-manager-producer match is called a *managed match*.

Direct match. Let $t \in \{1, \dots\}$ denote the period in a match. If a buyer i matches with a producer, the remainder of the period proceeds as follows. First, the buyer makes a payment $w_{it} \geq 0$ to the producer. The producer then chooses an effort level, denoted by $e_{it} \in \{0, 1\}$. The cost of effort is given by $c(e_{it})$, where $c(0) = 0$ and $c(1) = c > 0$. The effort generates an output for the buyer only if demand is positive, so $y_{it} = y d_{it} e_{it}$. Effort and output are observable by the buyer but are not verifiable by a court.

[4] See, e.g., Shapiro and Stiglitz (1984); MacLeod and Malcomson (1998); Board and Meyer-ter-Vehn (2015).

[5] Board and Meyer-ter-Vehn (2015) analyzes relational contracts in a frictionless matching market where matched producers may receive on-the-job offers. They show that this leads to heterogeneous productivity across otherwise identical matches. We rule this possibility out for simplicity.

Figure 2: Structure of direct and managed contracts



(a) Direct contract (b) Managed contract

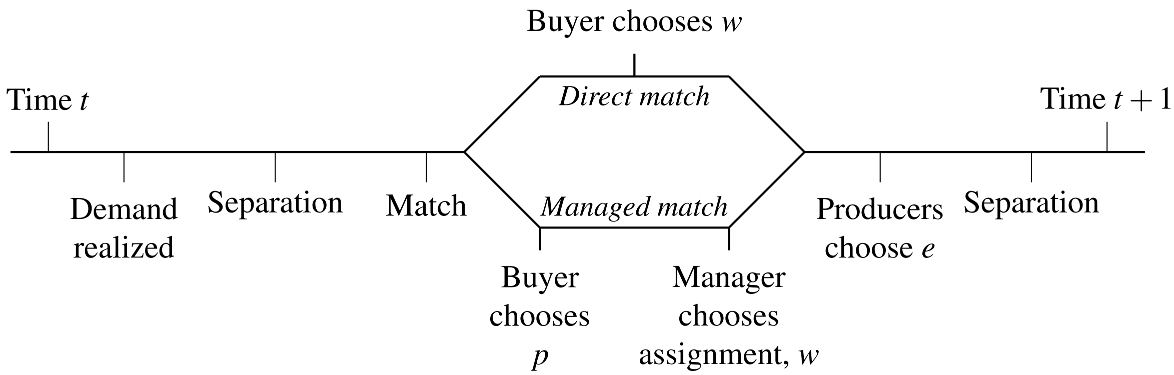
Managed match. If a buyer matches with a manager, she pays a service fee $p_{it} \geq 0$ to the manager. The manager chooses to assign one or none of its producers to the buyer. If no producer is assigned, the buyer receives no output. The manager pays w_{it} to the producer. The producer then exerts costly effort e_{it} and produces output y_{it} for the buyer. Effort and output are observable by both the buyer and the manager, but are not verifiable by a court.

Separation. Either party in a match may choose to separate from their match both after demand realization and after production. If separation occurs after demand realization, both parties can participate in the producer or manager market matching in the current period. However, if separation occurs after production, they need to wait till the next period to rematch.

Payoffs. In a direct match, the buyer's payoff is $y_{it} - w_{it}$ and the producer's is $w_{it} - c(e_{it})$ in each period t. In a managed match, the buyer's payoff is $y_{it} - p_{it}$, the manager's payoff is $p_{it} - w_{it}$, and the producer's payoff is $w_{it} - c(e_{it})$ in each period t.

Remark. This model has two key features that are not present in standard models of relational contracting in frictionless matching markets (e.g., Shapiro and Stiglitz (1984) and MacLeod and Malcomson (1998)). First, we allow buyers to have fluctuating demand. Second, we introduce managers who can enter relational contracts with a multitude of

Figure 3: Timeline of a single period



At the start of the period, buyer demand is realized and publicly observed. Separations may then occur. Unmatched buyers can open manager vacancies and match with managers. Unmatched buyers and managers can then open producer vacancies and match with unmatched producers. In a direct match, the buyer pays the producer and the producer exerts effort. In a managed match, the buyer pays the manager, the manager assigns a producer to the buyer and pays the producer, and the producer then exerts effort. Separation may occur at the end of the period.

players on both sides of the market. These managers do not have intrinsic demand or productive ability. What they can do, instead, is to match with a large set of buyers and producers, coordinate their matched producers, and monitor the producers' performance.[6] As we will show in Section 3 below, when demand is transient and the cost of idleness is high, managers can sustain cheaper relational contracts on both sides of the market by reassigning their producers across buyers. Buyers may therefore prefer managerial coordination to direct relational contracting.

2.1 Empirical Applications

Our model provides a lens to understand modern organizational forms that increasingly displace traditional decentralized exchange. The model's key friction—that effort is non-contractible and must be motivated by the future surplus of a relationship—is particularly

[6]We assume managers observe producer effort. In practice, this requires a communication channel from buyers to managers—for example, through customer reviews or performance reports. We abstract from the details of this communication, but it is implicitly present in any setting where the manager does not directly witness production. We also abstract from the possibility of strategic misreporting by buyers; in practice, repeated interactions or third-party verification may ensure the reliability of this channel.

salient in settings where quality is complex, performance is difficult to specify ex-ante, or legal enforcement is costly. The benefit of a manager is to aggregate fluctuating demand, allowing for longer, more stable producer relationships that ease this incentive problem. This logic applies naturally to the three examples below, which we present now to motivate the formal analysis.

Example 1 (Professional service firms)**.** In many professional labor markets (e.g., legal, accounting, consulting), the core service—such as providing expert advice or representation—is inherently a non-contractible effort problem. A client cannot perfectly specify in advance the quality of legal strategy or the thoroughness of an audit, nor can they easily verify it in court. Instead, they rely on the service provider's concern for future business to incentivize performance. The model applies to a client's decision to either hire a specialized lawyer directly or engage a large law firm.

A client with a short-lived legal need faces a problem: if they employ a lawyer directly, the lawyer knows the relationship will end after the case, creating a severe end-game problem that undermines the promise of future surplus to motivate effort. The manager—here, the law firm—solves this by pooling the demand from many such transient clients. It can promise the managed lawyer a steady stream of future cases, conditional on good performance today. This longer "shadow of the future" lowers the incentive rent required to motivate high-quality work. The friction is not that legal contracts are unenforceable, but that the effort margin most critical to quality is not. The firm's value, therefore, lies in its role as a stable, coordinating nexus that makes relational incentives sustainable for otherwise short-lived client demands.[7]

Example 2 (Supply chain intermediaries)**.** A core friction in global sourcing is the difficulty of contracting on manufacturer effort regarding product quality, especially across jurisdic-

[7] In the past half century, specialized, professional service firms have increasingly displaced traditional direct employees in providing a variety of services, including accounting, law, IT, security, logistics, and cleaning (Weil 2014). The literature on domestic outsourcing has argued that intermediary firms help clients overcome search frictions, redress adverse selection, circumvent wage-setting norms, and evade regulation (Autor 2009; Bernhardt et al. 2016). Few scholars, however, have carefully studied the possibility that intermediary firms may also strengthen performance incentives as centralized coordinators.

tions with weak legal enforceability. A retailer could attempt to contract directly with a network of overseas factories. However, if the retailer's product demand is volatile—a sudden spike for a new fashion item—the relationship with any one factory is intermittent. This intermittence weakens the factory's incentive to invest in non-contractible quality or to prioritize the retailer's urgent orders, as there is no reliable promise of future business.

A global sourcing firm (e.g., Li & Fung, Shein) acts as the manager. It aggregates orders from multiple, potentially fluctuating retailers, providing a stable and predictable workflow to its network of manufacturers. This stability allows the intermediary to build long-term relational contracts with producers, credibly threatening to withhold a significant stream of future business if quality slips. The contracting friction is thus two-fold: the difficulty of enforcing quality specifications across borders, and the demand transience that would make direct bilateral relationships unsustainable. The intermediary's coordination solves both, turning volatile aggregate demand into a stable stream of incentives for producers.[8]

Example 3 (Digital matching platforms)**.** Digital matching platforms (e.g., Uber, Airbnb) provide a clear application of our manager as a central coordinator. Consider a rider needing a ride and a driver with a car; a direct transaction is possible. The key friction is that the driver's effort—safe driving, politeness, clean car—is largely non-contractible. In a one-off spot transaction, the driver has no incentive to provide this effort beyond basic politeness. A platform like Uber, acting as the manager, aggregates demand from millions of riders, allowing it to offer drivers a stream of future trip requests.

The "manager" here doesn't direct drivers in the traditional sense but coordinates allocation through an algorithm and monitors performance through ratings. A driver who provides poor service (low effort) risks deactivation, losing access to the platform's entire stream of future riders. This is a powerful relational incentive enabled by the platform's

[8]Scholars increasingly document the importance of intermediaries in trade. Ahn et al. (2011) estimate that intermediaries facilitated 20% of China's exports in 2005, while Bernard et al. (2015) find that over 25% of Italian exporters act as intermediaries, accounting for 10% of national exports. The importance of relational contracts in international trade is also increasingly recognized (Macchiavello and Morjaria 2015, 2023; Antràs and Chor 2022). However, trade theory has largely focused on the role of intermediaries in overcoming search rather than trust frictions (e.g., Antras and Costinot 2011).

ability to pool demand. The model clarifies that the platform's value may not primarily lie in reducing search frictions (though that exists), but in its role as a central nexus that creates a sufficiently long-lived relationship with the driver to overcome the moral hazard inherent in each individual ride. The friction is most acute where the platform manages many short-term, anonymous transactions, making it the only feasible structure to enforce quality through the threat of exclusion from a future flow of work.[9]

3 Optimal Direct and Managed Relational Contracts

In this section we characterize optimal direct and managed relational contracts. We then analyze the buyer's choice between direct and managed relational contracts. This analysis explains why and when managed contracting may be preferred over direct contracting.

3.1 Direct Relational Contracts

We first focus on a single buyer who directly matches with a sequence of producers. The effect of other buyers is captured by the outside options available to the producers. We assume that the vacancies opened by other buyers are such that the pre-matching continuation value of an unmatched producer is stationary. Let $\overline{U} > 0$ denote the producer's pre-matching continuation value.

First, consider a buyer i in calendar time τ who is unmatched before producer matching. Suppose the buyer has demand and opens a vacancy with some continuation value U_i^1. Since matching is frictionless, the vacancy is filled with certainty if and only if $U_i^1 > \overline{U}$.

Next, consider a buyer who is matched with a producer at the start of period τ and they are in period t of the current match. Further suppose that they do not separate after demand

[9]The role of online platforms as providers of relational trust is emphasized by legal scholars such as Henderson and Churi (2019), but it has received less attention from economic theorists. Existing economic literature on online platforms largely focuses on platform pricing in the presence of usage and membership externalities (Rochet and Tirole 2003, 2006) and the impact of platforms on consumer search (Brynjolfsson and Smith 2000; Ellison and Ellison 2009). An exception is Hagiu and Wright (2015), who focus on the incentive effects of platforms, but instead model the organizational difference between vertical integration and multi-sided platforms as a difference in the allocation of decision rights, following Gibbons (2005).

realization. The rest of the period transpires as follows: first, the buyer pays its producer w, then the producer exerts effort e, and finally the producer or buyer can terminate the relationship.

Following Board and Meyer-ter-Vehn (2015), we say that strategies are *contract-specific* if (1) the current pay w_{it} only depends on past demand realizations, pay, and effort levels in the current match, $(d_{i1}, w_{i1}, e_{i1}, ..., d_{i,t-1}, w_{i,t-1}, e_{i,t-1}, d_{it})$, (2) current effort e_{it} additionally depends on w_{it}, (3) separation decisions at the end of the period additionally depend on e_{it}, and (4) separation decisions at the beginning of the next period additionally depend on $d_{i,t+1}$. Contract-specific strategies are thus independent of calendar time τ, the identity of the producer, or histories outside of the current match.

We define a *relational contract* as a subgame-perfect equilibrium in pure, contract-specific strategies where deviations from the equilibrium path are punished in the harshest possible way. In a direct match, this means that the producer shirks and quits if the buyer deviates from equilibrium pay w_{it}; similarly, the buyer fires the producer if he deviates from equilibrium effort e_{it}. For ease of notation, let w_{it}^d, e_{it}^d, and y_{it}^d denote pay, effort, and output when $d_{it} = d$.

The post-matching continuation payoff for a matched producer when $d_{it} = 1$ is:

$$U_{it}^1 = w_{it}^1 - c(e_{it}^1) + \delta \left[(1-\alpha_{1i}) U_{i,t+1}^1 + \alpha_{1i} \left(\beta_{i,t+1} U_{i,t+1}^0 + (1-\beta_{i,t+1}) \overline{U} \right) \right]. \tag{1}$$

Their post-matching continuation payoff when $d_{it} = 0$ is:

$$U_{it}^0 = w_{it}^0 - c(e_{it}^0) + \delta \left[\alpha_{0i} U_{i,t+1}^1 + (1-\alpha_{0i}) \left(\beta_{i,t+1} U_{i,t+1}^0 + (1-\beta_{i,t+1}) \overline{U} \right) \right], \tag{2}$$

where β_{it} indicates whether the buyer remains matched with the producer when $d_{it} = 0$.

The post-matching continuation payoff for the buyer when $d_{it} = 1$ is:

$$\Pi_{it}^1 = y_{it}^1 - w_{it}^1 + \delta \left[(1-\alpha_{1i}) \Pi_{i,t+1}^1 + \alpha_{1i} \left(\beta_{i,t+1} \Pi_{i,t+1}^0 + (1-\beta_{i,t+1}) \overline{\Pi}_i^0 \right) \right]. \tag{3}$$

Their post-matching continuation payoff when $d_{it} = 0$ is:

$$\Pi_{it}^0 = y_{it}^0 - w_{it}^0 + \delta \left[\alpha_{0i} \Pi_{i,t+1}^1 + (1 - \alpha_{0i}) \left(\beta_{i,t+1} \Pi_{i,t+1}^0 + (1 - \beta_{i,t+1}) \overline{\Pi}_i^0 \right) \right], \tag{4}$$

where $\overline{\Pi}_i^0$ and $\overline{\Pi}_i^1$ are the buyer's pre-matching continuation values when $d_{it} = 0$ and $d_{it} = 1$, respectively.

The relational contract must satisfy the following incentive constraints for the producer:

$$U_{it}^1 \geq w_{it}^1 + \delta \overline{U}, \forall t, \tag{P-IC-e1}$$

$$U_{it}^0 \geq w_{it}^0 + \delta \overline{U}, \forall t, \tag{P-IC-e0}$$

$$U_{it}^1 \geq \overline{U}, \forall t, \tag{P-IC1}$$

$$U_{it}^0 \geq \overline{U}, \forall t. \tag{P-IC0}$$

Constraints (P-IC-e1) and (P-IC-e0) require the producer to choose effort over shirking when effort is made on the equilibrium path, when the demand is 1 or 0, respectively. Constraints (P-IC1) and (P-IC0) require that the producer remain with the current buyer, when the demand is 1 or 0, respectively. If $\beta_{it} = 0$, equation (2) and constraints (P-IC-e0) and (P-IC0) do not apply, as the buyer immediately separates from the producer when demand is zero.

The relational contracts must also satisfy incentive constraints for the buyer:

$$\Pi_{it}^1 \geq \delta \left[(1 - \alpha_{1i}) \overline{\Pi}_i^1 + \alpha_{1i} \overline{\Pi}_i^0 \right], \tag{B-IC-w1}$$

$$\Pi_{it}^0 \geq \delta \left[(1 - \alpha_{0i}) \overline{\Pi}_i^0 + \alpha_{0i} \overline{\Pi}_i^1 \right], \tag{B-IC-w0}$$

$$\Pi_{it}^1 \geq \overline{\Pi}_i^1, \tag{B-IC1}$$

$$\Pi_{it}^0 \geq \overline{\Pi}_i^0. \tag{B-IC0}$$

Constraints (B-IC-w1) and (B-IC-w0) ensure that the buyer honors the payment to the producer.[10] Constraints (B-IC1) and (B-IC0) reflect the buyer's desire to retain the producer when there is a demand or not, respectively. As before, if $\beta_{it} = 0$, the term Π_{it}^0 and constraint (B-IC0) do not apply, as the buyer would immediately separate from the producer.

The buyer's problem is to choose a relational contract that maximizes profit Π_{i1}^1, subject to (B-IC-w1), (B-IC1), (P-IC-e1), and (P-IC1), as well as (B-IC-w0), (B-IC0), (P-IC-e0), and (P-IC0) whenever they apply.

Stationarity

We now show that we can restrict to *stationary* relational contracts—where strategies are time-invariant within a match, i.e., $e_{it}^1 = e_i^1$, $w_{it}^1 = w_i^1$, and $\beta_{it} = \beta_i$, for all t, and when applicable, $e_{it}^0 = e_i^0$ and $w_{it}^0 = w_i^0$ for all t.

Lemma 1. *Assume a stationary outside option $\overline{U} > 0$ and that a buyer-optimal relational contract exists. There exists a stationary relational contract that is also buyer-optimal.*

Proof. All omitted proofs are in the appendix. □

Intuitively, buyers would like to back-load payments to incentivize effort. Indeed, the optimal contract would be back-loaded if the buyer's outside option were exogenous and lower than the equilibrium value of a new match (as in a bilateral relationship without an external matching market). However, the outside option here is endogenously determined: if new producers can be hired on back-loaded terms, then the buyer would wish to separate from expensive incumbent producers and replace them with new ones. This replacement argument, originally due to Shapiro and Stiglitz (1984) and formalized by MacLeod and Malcomson (1989), implies that the buyer's outside option equals the value of a new match.

With constant demand, the replacement argument alone forces stationary pay. With fluctuating demand, the buyer must additionally choose whether to retain or separate from an idle producer. Lemma 1 shows that stationarity extends to this richer environment. The

[10] Since the producer exerts effort after the buyer makes the payment, the buyer would not receive the output should the buyer dishonor the payment to the producer. Therefore, there is no output or wage payment in period t on the right hand side of the constraint.

key observation, as in Levin (2003), is that the buyer's continuation problem at any period t is parameterized by the same time-invariant data: the primitives are constant, the producer outside option $\overline{U}$ is stationary, and the buyer's pre-matching values equal the value of the buyer's problem from scratch (by the replacement property). Since demand transitions are time-homogeneous Markov, the buyer's problem at t is identical to the problem at $t = 1$, so the optimal contract terms are time-invariant.[11]

Lemma 1 does not assert that the buyer-optimal contract is unique. Given any stationary optimum, one can construct non-stationary contracts—for instance, by setting pay and effort to zero in the first period. Such alternatives generate weakly lower buyer payoffs and are also weakly worse for producers, since they reduce early-period compensation without increasing later compensation. We focus on stationary contracts because, by Lemma 1, they are buyer-optimal, and because they are analytically tractable.

It can be shown that $e_i^1 = 1$ and $e_i^0 = 0$ in any buyer-optimal stationary relational contract (see Lemma A.1). Intuitively, effort in state $d = 0$ is pure waste: it costs the producer c but generates no output (since revenue requires positive demand), so the buyer bears the cost through higher compensation without any offsetting benefit. Effort in state $d = 1$ is the only source of revenue in the match. Without it, the relationship generates no surplus, and the buyer would strictly prefer to deviate by ending the match and starting a productive one with a new producer — making any non-productive contract self-defeating.

An important assumption is that the payment w_t is nonnegative. This *limited liability* assumption is made for two reasons. First, it is realistic in many real-world settings. Second, it greatly simplifies the analysis. Because of limited liability, equilibrium pay is always zero during periods when demand is zero (i.e., $w_i^0 = 0$, see Lemma A.2). One can thus characterize each buyer's choice between stationary direct and managed contracts simply by comparing the buyer's per-period payment when demand is one (see Lemma A.3).

Two assumptions are made to reflect the large, anonymous nature of the matching

[11]The time-invariance argument requires only that the primitives, the producer outside option $\overline{U}$, and the buyer's pre-matching values are constant, and that demand transitions are time-homogeneous. Limited liability ($w_t^0 \geq 0$, $w_t^1 \leq y$) is one such time-invariant constraint, but the result does not depend on it.

market. First, we assume that buyers and producers cannot observe each other's behavior in past relationships. This is important: if producer 2 could observe a buyer's behavior with producer 1, then 2 could punish the buyer for unfairly firing 1. This would lower the buyer's outside option (the value of rematching), relaxing the buyer's incentive constraints and thereby enabling the buyer to credibly promise back-loaded pay to producer 1. Second, we consider equilibria in contract-specific strategies. If the buyer and producers conditioned their behavior on calendar time or the identity of a producer, they could cut payments in the first period of the game for the first producers, without affecting incentives for subsequent producers. We rule out such strategies, since they require the first producer to believe they are special, which is inconsistent with our idea of a large anonymous matching market.

Finally, we disallow bonus payments, which play an important role in MacLeod and Malcomson (1998) and Levin (2003). In Appendix A.4, we show that this assumption is not critical. The proof extends the argument for a similar result in Board and Meyer-ter-Vehn (2015): For any contract with bonuses, there is a weakly more profitable contract for buyers without bonuses. Intuitively, the buyer prefers to pay the producer at the latest possible moment before e_t. It is therefore weakly profitable to shift the bonus b_{t-1} into the next period's expected pay, such that a proportionately increased amount is paid only if the producer is retained.

Optimal direct relational contract

Lemma 1 enables us to restrict our attention to buyer-optimal, contract-specific, stationary relational contracts. The time subscript is now dropped. By Lemmas A.1 and A.2, such contracts necessarily have $e_i^1 = 1$, $e_i^0 = 0$, and $w_i^0 = 0$. Let $\mathscr{C}_i^D = (w_i^1, \beta_i)$ denote such a contract, where w_i^1 is the payment when $d_{it} = 1$ and $\beta_i \in \{0,1\}$ indicates whether buyer i stays with the producer when demand is zero. The buyer-optimal direct contract is obtained by choosing w_i^1 and β_i to maximize Π_i^1, subject to (B-IC-w1), (B-IC1), (P-IC-e1), and (P-IC1), as well as (B-IC-w0), (B-IC0), (P-IC-e0), and (P-IC0) if $\beta_i = 1$.

Lemma 2. *Suppose a buyer-optimal direct relational contract exists. Under this relational* *contract, if*

$$\frac{(1-\delta)\overline{U}}{c} > \frac{\alpha_{0i}}{1-\alpha_{1i}}, \tag{5}$$

then the buyer pays

$$w_{DS}(\alpha_i) = \left(\frac{1}{\delta}\frac{1}{1-\alpha_{1i}}\right)c + (1-\delta)\overline{U}. \tag{6}$$

when demand is one and separates from the producer when demand switches to zero. Otherwise, the buyer pays

$$w_{DI}(\alpha_i) = \left(\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i})+\frac{\delta}{1-\delta}\alpha_{0i}}\right)c + \left(\frac{1+\frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i})+\frac{\delta}{1-\delta}\alpha_{0i}} - \delta\right)\overline{U} \tag{7}$$

when demand is one, remains matched with the idle producer when demand switches to zero, and pays zero until demand switches back to one.

Lemma 2 shows that the optimal direct contract features a buyer who either always separates from a producer or always retains him when their demand switches to zero. According to Equation (5), direct contracts with separation dominate direct contracts with idleness when (a) the producer's continuation value when unmatched $\overline{U}$ is high, (b) the buyer has a smaller α_{0i}, so a longer period of idleness is expected, and (c) the buyer has a larger α_{1i}, so a shorter period with service demand is expected.

The payment to producers in a direct contract, $w_D(\alpha_i) \equiv \min\{w_{DS}(\alpha_i), w_{DI}(\alpha_i)\}$, is increasing in α_{1i}. This is because when business needs are shorter-lived, the producer faces a higher chance of either separating or becoming idle, so the future surplus in the relationship is smaller. A higher incentive rent is therefore needed to incentivize producer effort.

3.2 Managed Relational Contracts

In managed matches, buyers enter relational contracts with managers, who are delegated the responsibility of motivating producers. The manager fulfills the buyer's demand by entering relational (sub-)contracts with a large number of producers and assigning producers to buyers according to demand realization. As we shall show, by aggregating fluctuating buyer demand and coordinating buyer-producer assignments, managers can reduce the cost of motivating producer performance.

To demonstrate this, we first analyze the (sub-)contracts that managers enter with producers. We argue here that unlike buyers, who have fluctuating demand, managers have constant demand for services. Why? In equilibrium, each manager randomly matches with a positive measure of buyers drawn from the same type distribution. Therefore, by the law of large numbers, the total measure of demand that managers receive from buyers is unchanging over time. Given this stable demand for services, the measure of producers that each manager contracts with is equal to the expected measure of demand realizations. Each matched producer in these contracts exerts effort in every period.

For tractability, we first assume that each manager enters manager-optimal, contract-specific, and stationary contracts with producers to meet buyer demand. This assumption can be justified by the same argument as in the previous subsection.

Second, we assume that producers are matched in a large anonymous market. This implies that managers will, with positive probability, immediately rematch with producers from whom they just separated, without knowing their histories. This assumption significantly increases analytic tractability, since it enables us to reuse the machinery developed for direct relational contracts to analyze subcontracts between managers and producers.

Importantly, this assumption is conservative: it yields an upper bound on the incentive pay required in managed contracts. If matching were not anonymous, a manager would never rematch with a producer it previously fired, and could therefore threaten more severe punishments for underperformance—since each manager contracts with a nonzero measure of producers. This would lower the incentive pay required in managed contracts, which is the mechanism highlighted by Biglaiser and Friedman (1994), where intermediaries offer lower incentive rents to sellers purely due to their market share.[12] Our assumption of anonymity shuts down this *market-share* benefit of managed contracts, allowing us to instead focus on

[12] Biglaiser and Friedman (1994) show middlemen with reputations have a larger incentive to monitor and are in a better position to learn about quality than an ordinary buyer does because they buy a larger proportion of the producers' goods (see also Bardhan et al. 2013). Unlike their model, managers do not have reputations in our baseline model; they simply enter relational contracts on both sides of the market.

the manager's *demand-smoothing* role in fluctuating environments.

Given these assumptions, the payment for the producer is $w_i^1 = w_M$ in every period. By the same logic as Lemma 2, we have that

$$w_M = \frac{c}{\delta} + (1-\delta)\overline{U}. \tag{8}$$

The demand-smoothing benefit of managed contracts is apparent from Equation (8). Since the managers can aggregate fluctuating demand across their buyers, their payments to producers (w_M) are lower than buyer payments to producers under direct contracts (w_D). Mathematically, $w_D(\alpha_i) \geq w_M$ for all α_i, where equality holds if and only if $\alpha_{1i} = 0$.

Next, we consider how producers are assigned to buyers in each period under the managed contract. Note that buyers are indifferent between any assignment of producers where the assigned producer exerts effort, since producers are identical in our model. Producers are also indifferent between any assignment of buyers where the managers offer the same level of payment. Since managers require producers to exert effort and provide the same compensating payment in every period, buyers and producers have the same payoffs in any assignment where producers are matched with buyers with positive demand in every period. There are an infinite number of such assignments, and any such assignment is optimal. This multiplicity gives rise to a need for each manager to coordinate assignments between their matched buyers and producers. For simplicity, we will not explicitly model the coordination process and instead assume that an optimal assignment is chosen.

Now, we characterize the contract between buyers and managers. The post-matching continuation payoffs for the manager, when the service is and is not needed, respectively, are

$$V_{it}^1 = p_{it}^1 - w_M + \delta\Big[(1-\alpha_{1i})V_i^1 + \alpha_{1i}(\beta_{it}V_{it}^0 + (1-\beta_{it})\overline{V})\Big], \tag{9}$$

and

$$V_{it}^0 = p_{it}^0 + \delta\Big[\alpha_{0i}V_{it}^1 + (1-\alpha_{0i})(\beta_{it}V_{it}^0 + (1-\beta_{it})\overline{V})\Big], \tag{10}$$

where $\overline{V}$ is a manager's continuation value after separating from a buyer. The relevant incentive constraints for the manager, similar with those for a producer, are

$$V_{it}^{1} \geq p_{it}^{1} + \delta\overline{V}, \tag{M-IC-w1}$$

$$V_{it}^{0} \geq p_{it}^{0} + \delta\overline{V}, \tag{M-IC-w0}$$

$$V_{it}^{1} \geq \overline{V}, \tag{M-IC1}$$

$$V_{it}^{0} \geq \overline{V}. \tag{M-IC0}$$

For the buyer, the continuation payoffs and incentive constraints are the same as in the direct contract, except that the payments w_i^0 and w_i^1 to the producer are replaced with service fee payments p_i^0 and p_i^1 to the manager. For concision, we omit these conditions, which simply repeat (B-IC-w1), (B-IC-w0), (B-IC1), and (B-IC0).

Unlike producer matches, we assume that matching between buyers and managers is *not* anonymous. Because of this, if a manager separates from a buyer, the manager cannot re-match with the initial buyer. The manager also cannot match with a new buyer, because all other potential buyers are matched with some manager and will not become unmatched on the equilibrium path. Therefore, $\overline{V} = 0$ on the equilibrium path.

The assumption of non-anonymous matching is important for our result that managed contracts are sometimes cheaper than direct contracts. Suppose instead that manager matching is anonymous. In this case, a buyer who separates from a manager will immediately rematch with the same manager with positive probability, but the buyer will not know the identity or history of this manager. A larger incentive rent will then be needed to motivate the manager.

Once again, we assume that buyers enter buyer-optimal, contract-specific, and stationary contracts with managers. Let $\mathscr{C}_i^M = (p_i^1, \beta_i)$ denote such a managed relational contract. Here p_i^1 is the service fee when the buyer needs the service. If the buyer deviates from the specified service fee, the manager does not assign a producer to the buyer and separates from the buyer. If instead the producer assigned by the manager deviates from the specified effort, the buyer separates from the manager after production. The optimal managed contract maximizes Π_i^1 subject to these incentive compatibility constraints.

Lemma 3. *Suppose an optimal managed contract exists. Under this contract, the buyer always retains the manager, pays zero service fees to the manager when there is no demand, and when there is demand, she pays the manager a service fee equal to*

$$p(\alpha_i) = \left(\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i})+\frac{\delta}{1-\delta}\alpha_{0i}}\right) w_M. \tag{11}$$

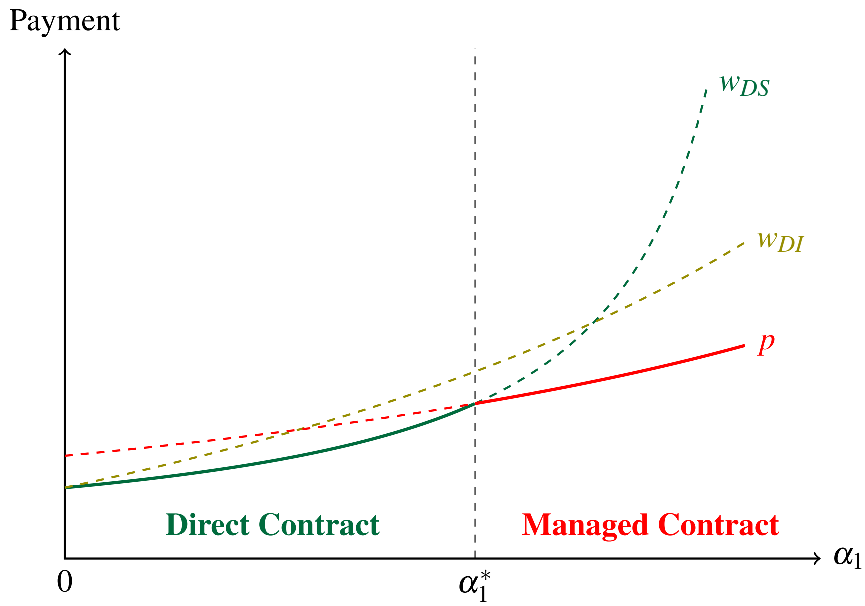
Lemma 3 shows that the cost of a managed contract is a form of double marginalization. Note that manager pay $p(\alpha_i)$ is the product of $\lambda(\alpha_i) = \frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i})+\frac{\delta}{1-\delta}\alpha_{0i}}$ and producer pay w_M. Here $\lambda(\alpha_i)$ can be thought of as a manager markup. Furthermore, as shown in Equation (8), w_M is elevated above the producer's cost of effort. In other words, both the manager and producer are paid rents so that both are incentivized to honor their contractual obligations.[13]

3.3 Optimal Contractual Choice

Having characterized direct and managed contracts, we now characterize when buyers choose managed contracts. We focus on buyers who have the same α_{0i}, which is the rate at which a buyer's demand changes from zero to one. We examine how the optimal contractual choice depends on α_{1i}, the rate at which the buyer's demand changes from one to zero, and $\overline{U}$, the producer's outside option, which in equilibrium reflects whether the producer market is tight and whether it is easy for producers to rematch.

[13] This form of double marginalization is different from the traditional textbook type. In the classic literature in industrial organization, double marginalization occurs when both upstream and downstream parties possess market power and use restrictive linear contracts. It can thus be eliminated by vertical integration of pricing decision rights, resale-price maintenance, or non-linear pricing (Tirole 1988). In our model, double marginalization instead arises from the non-verifiability of contract performance and the non-transferability of production and monitoring capabilities, as in models of middleman margins (e.g., Biglaiser and Friedman 1994; Bardhan et al. 2013). Conventional remedies do not eliminate this form of double marginalization.

Figure 4: Service cost under direct and managed relational contracts



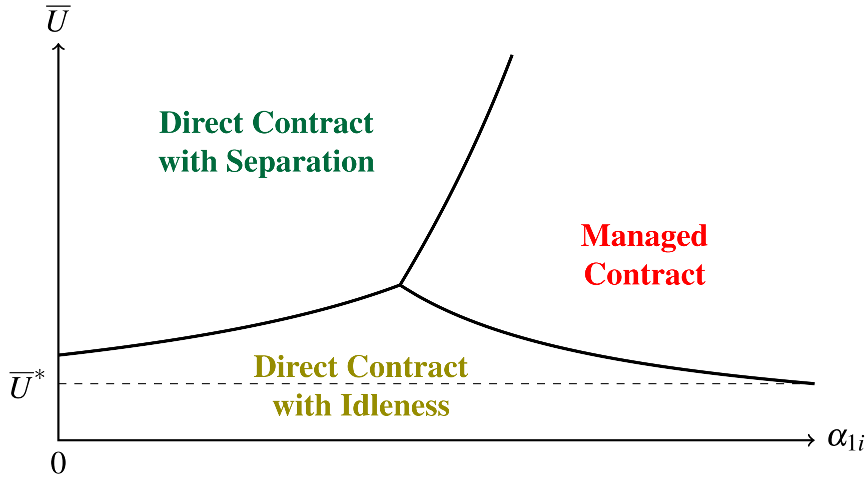
Note: α_1 is the probability that the buyer's demand switches from 1 to 0.

To compare the two, it suffices to compare the payment $w_D(\alpha_i)$ under direct contracting and the service fee $p(\alpha_i)$ under managed contracting. This is because the buyer pays nothing when their demand is zero and can always match with a producer when her demand is one.

Figure 4 provides a graphical illustration. For a buyer with stable demand, i.e., for whom $\alpha_{1i} = 0$, managed contracting is strictly more expensive than direct contracting because of double marginalization. To obtain high-quality services, the buyer needs to pay additional rent to the manager. However, there is no benefit to managed contracts, since demand is stable, so the producer is paid the same incentive rent under direct contracting.

The advantage of managed contracting over direct contracting becomes larger when business needs are short-term. As α_{1i} becomes larger, producers must be paid elevated payments in order for them to exert effort. Since managers can reassign producers across buyers, producers never separate or become idle, so the cost of incentivizing producers is lowered. To see this mathematically, note that $w_{DS}(\alpha_i)$ approaches infinity as α_{1i} approaches one. Furthermore, $p(\alpha_i)$ increases in α_{1i} less steeply than $w_{DI}(\alpha_i)$ if c is relatively small and $\overline{U}$ is relatively large. Therefore, when rematching is easy, the manager operates a more

Figure 5: Optimal contractual choice given model parameters



cost-efficient internal producer market by having long-standing relational contracts on both sides of the market.

Figure 5 graphically shows the optimal relational contracts as a function of α_{1i} and $\overline{U}$. In this figure, provided $\overline{U} > \overline{U}^*$, then there exists a cutoff value such that managed contracts dominate if and only if α_{1i} is sufficiently large. If $\overline{U} < \overline{U}^*$, then the producer's pre-matching continuation value is low, so it is optimal to retain them and keep them idle until demand returns.

The following proposition formalizes this result.

Proposition 1. *Take any set of buyers i with the same α_{0i} for whom an optimal contract exists. There exists $\overline{U}^* > 0$ such that:*

1. *If $\overline{U} > \overline{U}^*$, there exists $\alpha_1^*\left(\alpha_{0i}, \overline{U}\right) \in (0,1)$ such that a managed contract is optimal if and only if $\alpha_{1i} \geq \alpha_1^*(\alpha_{0i}, \overline{U})$;*

2. *If $\overline{U} < \overline{U}^*$, a direct contract is optimal for all i in this set.*

4 Steady-State Equilibrium

In this section, we show that there exists a unique steady-state equilibrium. We then explore the model's empirical implications. First, we compare the outcomes of managed producers and directly contracted producers in equilibrium. Second, we compare the outcomes and welfare of buyers and producers in economies with and without managers.

4.1 Deriving the Equilibrium

We say that the economy is in a steady state when (1) the number of producers in the economy and the distribution of contracts in the matching market are unchanging across periods and (2) each buyer's demand evolves according to its stationary distribution. By the properties of Markov processes, the steady-state probability that a buyer i has positive demand in any period is equal to $\pi_i = \alpha_{0i}/(\alpha_{0i}+\alpha_{1i})$.

At the start of each period, some buyers and managers are unmatched and open vacancies for unmatched producers. Let $\mathscr{I}_{DS}$ denote the set of buyers who enter direct contracts that end when their demand switches to zero. For a buyer $i \in \mathscr{I}_{DS}$, the steady-state probability that they offer new contracts is

$$v_i = (1-\pi_i)\alpha_{0i}. \tag{12}$$

Let $\mathscr{I}_{DI}$ be the set of buyers who directly contract with producers in contracts that continue when demand switches. Let $\mathscr{I}_M$ be the set of buyers who contract with managers. For both types of contracts, producers never separate. Therefore, for any buyer $i \in \mathscr{I}_{DI} \cup \mathscr{I}_M$, the steady-state probability that they open a vacancy is $v_i = 0$. The total measure of new vacancies opened by buyers and managers in the producer market is given by

$$v = \int_{\mathscr{I}_{DS}} v_i dF. \tag{13}$$

The steady-state measure of producers in direct contracts, n_B, is given by

$$n_B = \int_{\mathscr{I}_{DS}} \pi_i dF + \int_{\mathscr{I}_{DI}} dF. \tag{14}$$

The steady-state measure of producers in managed contracts, n_M, is given by[14]

$$n_M = \int_{\mathscr{I}_M} \pi_i dF. \tag{15}$$

The measure of unmatched producers after matching is

$$n_N = n - n_B - n_M. \tag{16}$$

If y and C are sufficiently large and the buyer type distribution F has full support on $[0,1] \times [0,1]$, the value of entering either direct or managed contracts is higher than the value of being unmatched. This implies that there is an excess of producers who enter the producer market, so $n_N > 0$.

Since matching is frictionless, all vacancies opened in the producer market are immediately filled. The total measure of unmatched producers before matching, u, is given by

$$u = v + n_N. \tag{17}$$

Matching is random and vacancies are never filled by producers that are already matched, so the Bellman equation for an unmatched producer is given by

$$\overline{U} = \int_{\mathscr{I}_{DS}} \frac{v_i}{u} U_{DS}(\alpha_i) dF + \left(1 - \frac{v}{u}\right) \overline{U}, \tag{18}$$

where $U_{DS}(\alpha_i) = \frac{1}{1-\delta(1-\alpha_{1i})}(w_{DS}(\alpha_i) - c + \delta\alpha_{1i}\overline{U})$. Note that the continuation value of being unmatched ($\overline{U}$) consists of two components. The first term reflects the rent from matching with an unmatched buyer. The second term reflects the value of remaining

[14] Here π_i enters the integral since the manager aggregates the fluctuating demand of buyers.

unmatched.

To complete the model, we solve for the steady-state number of producers that enter the economy. By assumption, producers enter the economy at the beginning of each period by paying an entry cost C. Entry drives down the likelihood that producers are matched, so they enter only until the continuation value of being unmatched in the labor market equals their entry cost. This yields the following condition:

$$\overline{U} = C. \tag{19}$$

We can now define an equilibrium in our economy.

Definition 1. *A steady-state equilibrium is a distribution of direct and managed relational contracts for buyers such that:*

1. *All relational contracts are buyer-optimal, contract-specific, and stationary;*
2. *Each player's pre-matching continuation value is determined by steady-state transition probabilities and frictionless and random matching via Equation* (18)*;*
3. *The measure of producers in the economy is derived from the producer entry condition, given by Equation* (19)*.*

Proposition 2. *There exists a unique steady-state equilibrium. If y and C are sufficiently high and F has full support on $[0,1]\times[0,1]$, then a non-zero measure of buyers enter direct contracts, while another non-zero measure of buyers enter managed contracts.*

Proof. By Equation (19), $\overline{U}$ equals the entry cost C. Given $\overline{U}$, we can compare the values of $w_{DS}(\alpha_i)$, $w_{DI}(\alpha_i)$, $p(\alpha_i)$, and y for each i using Equations (6), (7), and (11) to determine $\mathscr{I}_{DS}$, $\mathscr{I}_{DI}$, and $\mathscr{I}_M$. Having derived these, we can obtain unique values for v_i, v, n_B, and n_M from Equations (12), (13), (14), and (15). We plug these into Equation (18) to solve for a unique value for u. Plugging u into Equations (16) and (17) then yields a unique value for n. The desired statement then follows from Proposition 1. $\square$

4.2 Empirical Implications

Having shown that there exists a unique steady-state equilibrium in this exchange economy, we can now explore the model's predictions regarding equilibrium behavior.

Effects of managerial coordination on producers

How does managerial coordination affect the pay, separation rates, and idleness of producers? Recent research shows that workers managed by professional service firms earn lower and more compressed wages (Dube and Kaplan 2010; Goldschmidt and Schmieder 2017; Drenik et al. 2023). Some evidence also suggests that they have lower hazard into unemployment than comparable direct employees (Guo et al. 2025).

Similar patterns are found in our model. To see this, consider the interesting case where all three contract types arise in equilibrium.[15] We define a producer's *separation rate* as the probability that a matched producer becomes unmatched at the start of the next period, and *idleness* as the steady-state probability that a producer is matched but does not exert effort.

Corollary 1. *Directly contracted producers have (1) higher average pay, (2) more dispersed pay, (3) higher separation rates, and (4) higher idleness than managed producers.*

The first result follows from the fact that $w_D(\alpha_i) > w_M$ for all buyers i with fluctuating demand. The second follows from the fact that $w_D(\alpha_i)$ takes on different values depending on α_i, which is heterogeneous across buyers, while w_M is constant. The final two follow from the fact that directly contracted producers may separate from buyers or become idle when demand changes, while managed producers are always reallocated among the manager's clients, so they never separate nor become idle.

Trade and welfare with and without managerial coordination

How does the presence of managers alter market equilibrium and the welfare of buyers and producers? Felix and Wong (2025) show that Brazil's 1993 legalization of professionally

[15] Specifically, we assume that y, C and F are such that $|\mathscr{I}_M|, |\mathscr{I}_{DI}|, |\mathscr{I}_{DS}| > 0$. If F has full support on $[0,1] \times [0,1]$, then by Proposition 1 there exists y and C such that this holds.

managed security firms persistently increased security guard employment, but displaced incumbent security guards from high-wage direct employers.

To explore the same question theoretically, we consider two economies: one with managers and one without. The introduction of managers causes three types of buyers to switch to managed contracts: buyers who do not initially consume services, buyers who initially choose direct contracts with idleness, and buyers who initially choose direct contracts with separation. We denote the three subsets of switching buyers as $\mathscr{S}_0$, $\mathscr{S}_I$, and $\mathscr{S}_S$. We assume that there is a positive measure of switchers who initially do not consume, and that there is a positive measure of switchers who directly contract initially.[16]

In both economies, producers are assumed to freely enter at cost C. As such, the introduction of managers does not alter the payments received by producers who remain directly matched to the same buyers. It only reduces the payments received by producers who become matched to managers instead of buyers. It follows that:

Corollary 2. *When managers are introduced into an economy, the payoffs of all buyers weakly increase and the measure of buyers who receive services increases. The mean pay per unit effort received by producers falls. The total measure of matched producers increases if and only if* $\int_{\mathscr{S}_0} \pi_i dF > \int_{\mathscr{S}_I} (1-\pi_i) dF$.

These predictions are intuitive. The introduction of managers benefits buyers with fluctuating demand. As such, it unambiguously increases the measure of buyers who receive services. However, it hurts incumbent producers who were initially earning pay premiums. It is therefore ambiguous whether the measure of matched producers ($n_M + n_B$) increases or falls. On one hand, managers enable more buyers to afford services, which increases demand for producers. On the other, managers reduce idleness, so fewer producers are needed. The relative magnitude of these two effects depends on the distribution of buyer types.

[16]In other words, y, C and F are such that $|\mathscr{S}_0| > 0$ and $|\mathscr{S}_I \cup \mathscr{S}_S| > 0$. Assuming indifferent buyers do not switch, this requires that there exists $\alpha_i, \alpha_i' \in \operatorname{supp} F$ such that $y > w_D(\alpha_i) > p(\alpha_i)$ and $w_D(\alpha_i') > y > p(\alpha_i')$. By Proposition 1, if F has full support on $[0,1] \times [0,1]$, then there exist y and C such that this holds.

5 Extensions

This section extends the model to study the relationship between managerial coordination and producer specialization. We show that buyers sort into managerial coordination when both demand is transient and needs are specialized. We also characterize how managerial coordination impacts the division of labor. A further extension with an exogenous producer population is presented in the Appendix and briefly discussed below.

5.1 Producer Specialization

We consider a multi-task economy with a measure of buyers who have unit demand in every period, for either one of two tasks $d_{it} \in \{A, B\}$. Each buyer's demand switches from one task to another task with some symmetric probability α_i at the start of each period.[17] There is also an excess measure of producers indexed by j who choose whether to become either specialists in one of two activities needed by buyers or a generalist with middling skill in both services. Let $\phi_j \in \{A, B, G\}$ denote the chosen type of the producer, where A and B refer to specialists and G refers to the generalist. The output depends on the buyer's demand d_{it}, the producer's type ϕ_j, and the producer's chosen effort e_{jt}, and is given by

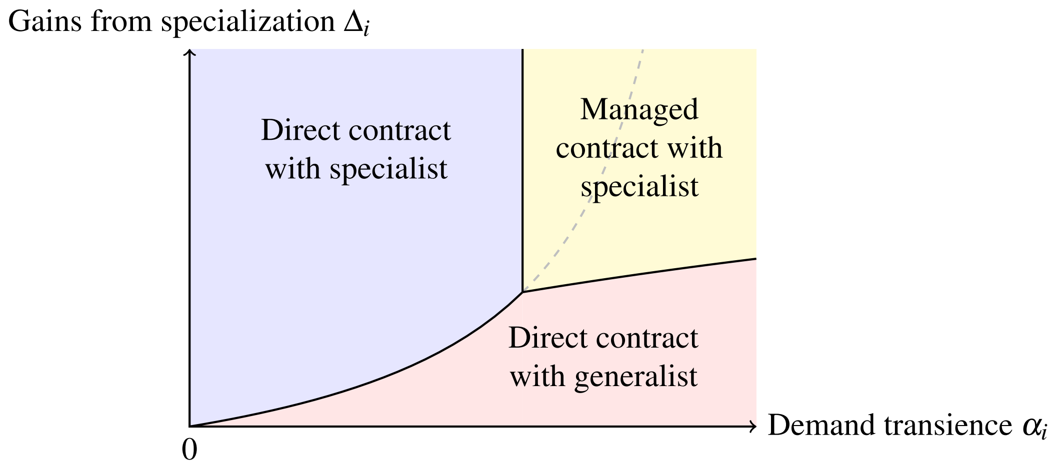
$$y_{ijt} = \begin{cases} y + \Delta_i, & \text{if } \phi_j = d_{it} \text{ and } e_{jt} = 1 \\ y, & \text{if } \phi_j = G \text{ and } e_{jt} = 1 \\ 0, & \text{otherwise} \end{cases}$$

where $\Delta_i > 0$ denotes the buyer-specific *gains from specialization*. Output is high when the producer is specialized, demand and the producer's type are well-matched, and effort is high. Output is middling for generalists who exert effort, regardless of demand. Output is low if either effort is not made or demand is misaligned with the chosen specialist.

As before, neither pay w nor effort e are contractible and must be incentivized through

[17] This is essentially simplifying the model in Section 2 by assuming $\alpha_{Ai} = \alpha_{Bi} = \alpha_i$, which reduces the type space to one dimension.

Figure 6: Optimal contractual choice in the presence of gains from specialization



relational contracts. We assume that output y is sufficiently large so that buyers are always able to receive positive profit by directly contracting with a generalist, so there are never buyers who do not receive services. We also assume that each producer's entry cost C is sufficiently large so that specialist producers never remain in a contract but become idle when the demand of their buyer changes. These assumptions allow us to focus on each buyer's choice between direct contracting with a generalist, direct contracting with a specialist who is never idle, and managed contracting. There are K managers for each task. Each manager can only monitor producers specializing in that task and are randomly matched with buyers who offer managed contracts.[18]

Drivers of managerial coordination in a multi-task economy

Abraham and Taylor (1996) find that establishments with cyclical demand and specialized needs are more likely to outsource accounting services to professionally managed firms. Similar patterns are found in our model, as visualized in Figure 6 and stated formally below.

Proposition 3. *A unique steady-state equilibrium exists in a multi-task economy. Buyer i chooses managed contracts if and only if their demand transience α_i and gains from specialization Δ_i are both sufficiently large.*

[18] This setup implicitly assumes that each manager specializes in monitoring one type of task.

Division of labor with and without managerial coordination

How does managerial coordination affect the division of labor and matching market dynamics? Correlational evidence shows that the rise of professional service firms has coincided with increasingly specialized workers and firms as well as falling unemployment (Katz and Krueger 1999; Song et al. 2019; Handwerker 2023).

To explore the same question theoretically, we compare multi-task economies with and without managers. We assume that the underlying parameters are such that a non-zero measure of buyers choose managed contracts if managers are present and that the conditional distribution of α_i given Δ_i has positive support on $[0, 1]$.

Corollary 3. *When managers are introduced into a multi-task economy, the measure of specialist producers increases and the measure of unmatched producers falls.*

The intuition is as follows. The dashed curve in Figure 6 shows the boundary between direct contracting with a specialist and with a generalist when managers are absent. When managers are present, part of the demand for directly contracted generalists is replaced with demand for managed specialists. Overall demand for specialists rises. In response, more producers choose to become specialists. Correspondingly, there are fewer direct contracts with elevated pay, so fewer producers enter, and the measure of unmatched producers falls.

5.2 Exogenous Producer Population

Appendix C considers a version of the baseline model where the number of producers $n > 1$ is fixed rather than determined by entry. With a fixed producer pool, managers affect the producer outside option $\overline{U}$ by changing aggregate demand for producers, so equilibrium wages and contract choices adjust endogenously. The appendix shows there is a unique steady-state equilibrium and that, if n is large enough, the market becomes very slack, $\overline{U}$ approaches zero, buyers optimally use direct contracts with idleness, and managerial coordination disappears. Moreover, when managers are introduced with fixed n, they have both direct effects (buyers switching to managed contracts and lowering producer rents) and indirect effects through the induced change in $\overline{U}$, which can raise or lower all producers' pay depending on how many previously unmatched buyers become matched.

6 Conclusion

Economists have long puzzled over why the *visible hand* of managers often replaces the *invisible hand* of markets in allocating resources. This paper sheds new light on the puzzle by developing a repeated-game model of an exchange economy in which managers coordinate allocations in decentralized matching markets. We show that managerial coordination reduces the cost of incentivizing performance in fluctuating environments. However, managerial coordination involves a trade-off between demand smoothing and double marginalization. Consequently, managers only partially replace anonymous matching markets in determining assignments between buyers and producers. In equilibrium, buyers sort into managerial coordination and decentralized exchange depending on their demand transience and their gains from specialization. Managerial coordination increases trade and promotes specialization, but redistributes surplus from producers to buyers.

A key contribution of our work is to show that managerial coordination is surplus-enhancing in exchange economies, even in the absence of many factors previously hypothesized to be important, including contract-writing costs, bargaining frictions, search frictions, and adverse selection. The predictions of the model are applicable to a variety of organizational forms. They are also consistent with recent empirical findings on professional service firms. Promising directions for future work include heterogeneous managers, endogenous monitoring technology, and richer information structures, where the interplay between relational incentives and market structure may yield further insights into the organization of economic activity.

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Appendix

A Model with Endogenous Producer Entry

A.1 Proof of Stationarity, Lemma 1

We omit the subscript i. A contract is a sequence $(e_t^1, e_t^0, w_t^1, w_t^0, \beta_t)_{t\geq 1}$ adapted to the demand history, where $e_t^d \in \{0,1\}$ is the prescribed effort when $d_t = d$, w_t^d is the payment when $d_t = d$, and $\beta_t \in \{0,1\}$ indicates whether the buyer retains the producer when $d_t = 0$. When $e_t^d = 1$, the producer exerts effort at cost c and produces output y if $d_t = 1$ (and zero output if $d_t = 0$); when $e_t^d = 0$, the producer exerts no effort and produces nothing. The buyer's problem is

$$\sup_{(e_t^1, e_t^0, w_t^1, w_t^0, \beta_t)_{t\geq 1}} \Pi_1^1$$

subject to (P-IC-e1), (P-IC-e0), (P-IC1), (P-IC0), (B-IC-w1), (B-IC-w0), (B-IC1), (B-IC0),

where the constraints hold for all $t \geq 1$ and in each demand state for which they are relevant.

Step 1: Replacement property and time-invariance. Let V^d denote the supremum of the buyer's problem starting from period 1 in demand state d. We establish that $\overline{\Pi}^d = V^d$ for each $d \in \{0,1\}$.

First, $\overline{\Pi}^d \geq V^d$. Matching is frictionless with an excess supply of producers, so the buyer can always fill a vacancy. At the start of any period with demand state d, the buyer can separate from the incumbent, hire a new producer with outside option $\overline{U}$, and solve the buyer's problem from scratch. Since all producers are homogeneous, the resulting payoff is V^d. Because separation is always available, the buyer's equilibrium payoff—and hence the pre-matching continuation value $\overline{\Pi}^d$—is at least V^d.

Second, $\overline{\Pi}^d \leq V^d$. The pre-matching continuation value $\overline{\Pi}^d$ is the buyer's equilibrium payoff upon entering the matching market in state d, which is attained by some feasible contract-specific relational contract. Since V^d is the supremum over all feasible contract-specific relational contracts starting in state d, we have $\overline{\Pi}^d \leq V^d$.

Combining,

$$\overline{\Pi}^1 = V^1, \qquad \overline{\Pi}^0 = V^0. \tag{20}$$

Now fix any $t \geq 1$ and $d \in \{0,1\}$. The buyer's continuation problem from period t onward, conditional on $d_t = d$, is to choose $(e_s^1, e_s^0, w_s^1, w_s^0, \beta_s)_{s \geq t}$ to maximize Π_t^d subject to the incentive constraints (P-IC-e1)–(B-IC0) for all $s \geq t$. The generalized Bellman equations define Π_t^d and U_t^d recursively from period t onward. We verify that every object entering the recursive definition and the constraints is independent of t:

(i) the primitives $(\alpha_0, \alpha_1, y, c, \delta)$ and the producer outside option $\overline{U}$ are constant;

(ii) the buyer's pre-matching values $(\overline{\Pi}^0, \overline{\Pi}^1) = (V^0, V^1)$ are constant by the replacement property (20);

(iii) the demand process is Markov with time-homogeneous transition probabilities, so the distribution of future demand paths conditional on $d_t = d$ does not depend on t.

Since neither the objective nor the constraint set depends on t, the continuation problem at (t,d) is identical to the problem at $(1,d)$. Its value is therefore V^d, independent of t.

Step 2: Construction of a stationary optimum. By Step 1, the buyer's continuation problem at any $(t, d_t = d)$ is identical to the problem at $(1,d)$, so its solution depends only on the current demand state d. Let $(\hat{e}^1, \hat{e}^0, \hat{w}^1, \hat{w}^0, \hat{\beta})$ denote a solution to the buyer's problem at period 1 with $d_1 = 1$, taking as given that the continuation values from period 2 onward are (V^0, V^1). Such a solution exists by assumption; by Step 1, the period-1 actions in any buyer-optimal contract solve this problem. Define the stationary contract by setting

$$(e_t^1,\, e_t^0,\, w_t^1,\, w_t^0,\, \beta_t) \;=\; (\hat{e}^1,\, \hat{e}^0,\, \hat{w}^1,\, \hat{w}^0,\, \hat{\beta}) \qquad \text{for all } t \geq 1.$$

We verify two properties.

- *Continuation values equal* (V^0, V^1). Under the stationary contract, the generalized Bellman equations have constant coefficients: the actions $(\hat{e}^1, \hat{e}^0, \hat{w}^1, \hat{w}^0, \hat{\beta})$, the outside options $(\overline{\Pi}^0, \overline{\Pi}^1) = (V^0, V^1)$, and the primitives are all time-invariant. The continuation values Π_t^d therefore satisfy the same fixed-point equations at every t and are constant; write them as Π^d. We claim $(\Pi^0, \Pi^1) = (V^0, V^1)$. To see this, note that when $(\hat{e}^1, \hat{e}^0, \hat{w}^1, \hat{w}^0, \hat{\beta})$ was chosen to solve the buyer's problem at $(1, d=1)$ with future continuation values (V^0, V^1), the resulting period-1 payoff satisfied the Bellman equations with $\Pi_{t+1}^d = V^d$. But these are exactly the same Bellman equations that the stationary continuation values (Π^0, Π^1) must satisfy. Since $\delta < 1$, the linear system has a unique solution, so $\Pi^d = V^d$. [19]

- *Feasibility.* Since $(\Pi^0, \Pi^1) = (V^0, V^1)$, the constraints (P-IC-e1)–(B-IC0) at any (t, d) under the stationary contract are identical to those at $(1, d)$ with continuation values (V^0, V^1). The actions $(\hat{e}^1, \hat{e}^0, \hat{w}^1, \hat{w}^0, \hat{\beta})$ satisfy these constraints at $(1, d)$ by construction, so they are satisfied at every t.

Step 3: Optimality. By definition, V^1 is the supremum of Π_1^1 over all feasible contract-specific relational contracts. The stationary contract constructed in Step 2 attains this supremum ($\Pi_1^1 = V^1$), so it is optimal. In particular, for any other feasible contract, the stationary contract yields weakly higher buyer profit. □

A.2 Optimal Effort Rule

Lemma A.1. *In any buyer-optimal, contract-specific, stationary relational contract, $e_i^1 = 1$ and $e_i^0 = 0$.*

Proof. First, consider e^0. When $d_t = 0$, output is zero regardless of effort. Take any feasible stationary contract with $e^0 = 1$ and modify it by setting $e^0 = 0$ and $\hat{w}^0 = \max\{w^0 - c, 0\}$.

[19] The generalized Bellman equations with constant coefficients can be written as $x = Ax + b$ where A is the discounted transition matrix. Since $\delta < 1$, the spectral radius of A is strictly less than 1, guaranteeing a unique fixed point.

The producer's per-period flow in state $d = 0$ changes from $w^0 - c$ to $\hat{w}^0 \geq w^0 - c$, so U^0 weakly increases. Since U^0 enters the incentive constraints for state $d = 1$ only through the retention value (which is weakly higher), all constraints are weakly relaxed. Meanwhile, the buyer's per-period flow in state $d = 0$ changes from $-w^0$ to $-\hat{w}^0 \geq -w^0$, so the buyer is weakly better off in every period. Therefore $e^0 = 0$ weakly dominates $e^0 = 1$.

Second, consider e^1. If $e^1 = 0$, then (since $e^0 = 0$) no output is generated in either demand state. We claim $\Pi^1 < V^1$ for any such contract. The best case for the buyer is $w^1 = w^0 = 0$ and $\beta = 0$, giving $\Pi^1 = \delta[(1-\alpha_1)\Pi^1 + \alpha_1 V^0]$, while the optimal productive contract satisfies $V^1 = (y - w^{1*}) + \delta[(1-\alpha_1)V^1 + \alpha_1 V^0]$ with $y - w^{1*} > 0$. Subtracting yields $(1-\delta(1-\alpha_1))(\Pi^1 - V^1) = -(y - w^{1*}) < 0$, so $\Pi^1 < V^1$. But the buyer's incentive constraints require $\Pi^1 \geq \overline{\Pi}^1 = V^1$, since the buyer can always deviate by ending the current match and starting a new, productive one. This constraint is violated, so any contract with $e^1 = 0$ is infeasible when $V^1 > 0$. □

A.3 Implications of Limited Liability

We establish the following two results.

Lemma A.2. *Suppose a buyer-optimal, contract-specific, stationary relational contract exists. Under this contract, if $\beta = 1$, then the following three conditions hold: 1) $w^0 = 0$, 2) both (B-IC1) and (B-IC0) bind, and 3) (P-IC1) and (P-IC-e0) are slack.*

Proof. Suppose for contradiction that $w^0 > 0$. Then $\Pi^0 = -w^0 + \delta[\alpha_0 \Pi^1 + (1-\alpha_0)\Pi^0]$ and $\overline{\Pi}^0 = \delta[\alpha_0 \overline{\Pi}^1 + (1-\alpha_0)\overline{\Pi}^0]$. Using $\Pi^1 = \overline{\Pi}^1$ and $\Pi^0 = \overline{\Pi}^0$ from the replacement property (20), we obtain $\Pi^0 - \overline{\Pi}^0 = -w^0 + \delta(1-\alpha_0)(\Pi^0 - \overline{\Pi}^0)$, which implies $\Pi^0 - \overline{\Pi}^0 = -w^0/(1-\delta(1-\alpha_0)) < 0$, contradicting $\Pi^0 = \overline{\Pi}^0$. Hence $w^0 = 0$. Given (P-IC0), this in turn implies that (P-IC-e0) is slack. The replacement property implies that both (B-IC1) and (B-IC0) bind.

We now show that (P-IC1) is slack. Suppose it binds, so $U^1 = \overline{U}$. Then given $w^0 = 0$ and $\delta < 1$, plug in $U^1 = \overline{U}$ and get $U^0 = \delta\left[\alpha_0 \overline{U} + (1-\alpha_0)(\beta U^0 + (1-\beta)\overline{U})\right] < (1-$

$\alpha_0)\beta U^0 + (1-(1-\alpha_0)\beta)\overline{U}$. This inequality implies that $U^0 < \overline{U}$, which contradicts (P-IC0). So (P-IC1) is slack. $\square$

Lemma A.3. *Maximizing Π^1 among buyer-optimal, contract-specific, stationary relational contracts is equivalent to minimizing w^1.*

Proof. The lemma is obvious in the case where $\beta = 0$. If $\beta = 1$, by Lemma A.2, $\Pi^0 = \frac{\delta\alpha_0}{1-\delta(1-\alpha_0)}\Pi^1$. Substituting into the Π^1 Bellman, using Lemma A.1, and solving:

$$\Pi^1 = \frac{1-\delta(1-\alpha_0)}{(1-\delta)\big[1-\delta(1-\alpha_0-\alpha_1)\big]}\,(y-w^1).$$

Since $\delta < 1$ and $\alpha_0, \alpha_1 \in (0,1)$, the coefficient is strictly positive, so $\partial\Pi^1/\partial w^1 < 0$. $\square$

A.4 Irrelevance of bonus payments

We extend the logic of (Board and Meyer-ter-Vehn, 2015, Appendix A.2) to an environment with fluctuating demand and retention decisions. The key insight is that the buyer prefers to compensate the producer via the pay rather than a post-effort bonus, because this maximizes the producer's continuation value at risk at the moment of the effort decision.

Step 1: Setup. Consider a model in which, after the producer exerts effort e_t, the buyer may pay a bonus $b_t^d \geq 0$. The continuation payoffs are:

$$U_t^1 = w_t^1 - c(e_t) + b_t^1 + \delta\big[(1-\alpha_1)\,U_{t+1}^1 + \alpha_1\big(\beta_t\,U_{t+1}^0 + (1-\beta_t)\,\overline{U}\big)\big], \tag{21}$$

$$U_t^0 = w_t^0 + b_t^0 + \delta\big[\alpha_0\,U_{t+1}^1 + (1-\alpha_0)\big(\beta_t\,U_{t+1}^0 + (1-\beta_t)\,\overline{U}\big)\big], \tag{22}$$

with analogous expressions for Π_t^1 and Π_t^0. In addition to the constraints in the main text, the buyer must honor the bonus after observing effort:

$$-b_t^1+\delta\big[(1-\alpha_1)\,\Pi_{t+1}^1+\alpha_1\big(\beta_t\,\Pi_{t+1}^0+(1-\beta_t)\,\overline{\Pi}^0\big)\big]\geq\delta\big[(1-\alpha_1)\,\overline{\Pi}^1+\alpha_1\,\overline{\Pi}^0\big], \tag{B-IC-b1}$$

$$-b_t^0+\delta\big[\alpha_0\,\Pi_{t+1}^1+(1-\alpha_0)\big(\beta_t\,\Pi_{t+1}^0+(1-\beta_t)\,\overline{\Pi}^0\big)\big]\geq\delta\big[\alpha_0\,\overline{\Pi}^1+(1-\alpha_0)\,\overline{\Pi}^0\big]. \tag{B-IC-b0}$$

Step 2: Shifting bonuses into next-period pay. Fix any feasible contract $(e_t,w_t^1,b_t^1,w_t^0,b_t^0,\beta_t)_t$. We construct a new contract without bonuses by absorbing each bonus into the next period's pay. Set $\hat{e}_t=e_t$, $\hat{\beta}_t=\beta_t$, and

$$\hat{w}_{t+1}^{d,1}=w_{t+1}^d+\frac{b_t^1}{\delta\big[(1-\alpha_1)+\alpha_1\,\beta_{t+1}\big]}, \tag{23}$$

$$\hat{w}_{t+1}^{d,0}=w_{t+1}^d+\frac{b_t^0}{\delta\big[\alpha_0+(1-\alpha_0)\,\beta_{t+1}\big]}, \tag{24}$$

where $\hat{w}_t^{d,d'}$ denotes the period-t pay conditional on current demand d and previous-period demand d'. The bonus $b_t^{d'}$ is spread across exactly the states where the match continues into $t+1$, so the buyer's expected cost is unchanged. Note that the denominators in (23) and (24) are strictly positive whenever $\alpha_1<1$ and $\alpha_0>0$ respectively, which holds by assumption.

Step 3: Verification. Since the bonus is merely prepaid as a pay, the buyer's continuation payoffs $\hat{\Pi}_t^d$ are identical to Π_t^d for all t. Constraints (B-IC-b1)–(B-IC-b0) become vacuous, and (B-IC-w1), (B-IC-w0) are preserved because buyer payoffs are unchanged.

For the producer, shifting the bonus into the pay (weakly) *strengthens* incentives. Under the original contract, (P-IC-e1) requires

$$b_t^1+\delta\big[(1-\alpha_1)(U_{t+1}^1-\overline{U})+\alpha_1\,\beta_t\,(U_{t+1}^0-\overline{U})\big]\geq c.$$

In the new contract, $\hat{b}_t^1=0$ but the higher pays raise continuation values: $\hat{U}_{t+1}^d\geq U_{t+1}^d$ for each d and t, because the producer receives the shifted bonus amount as part of her period-$(t+1)$ pay regardless of her effort choice. Since the bonus b_t^1 was paid only after effort was exerted, the producer could not forfeit it by shirking. By contrast, the higher pay $\hat{w}_{t+1}^{d,1}$ is received at the start of period $t+1$ and is therefore part of the continuation value that the producer puts at risk when deciding whether to exert effort. Formally, the new (P-IC-e1) becomes

$$\delta\left[(1-\alpha_1)(\hat{U}_{t+1}^1-\overline{U})+\alpha_1\beta_t(\hat{U}_{t+1}^0-\overline{U})\right]\geq c,$$

which is satisfied because $\hat{U}_{t+1}^d-\overline{U}\geq U_{t+1}^d-\overline{U}$ and the original (P-IC-e1) held with the bonus $b_t^1\geq 0$ included. A similar logic holds for (P-IC-e0). Retention constraints (P-IC1) and (P-IC0) are similarly (weakly) relaxed, and non-negativity of pay follows from $b_t^d\geq 0$ and $w_{t+1}^d\geq 0$.

The new contract therefore generates the same buyer payoff while satisfying all constraints, so it weakly dominates the original. The resulting contract may depend on the previous period's demand state through $\hat{w}_t^{d,d'}$, but since it can then be made stationary by Lemma 1, this dependence is eliminated in the buyer-optimal contract.

A.5 Proof of Optimal Direct Contract, Lemma 2

For simplicity, we write w^1 as w in the rest of the proof. We complete the proof by analyzing and comparing the terms in the buyer-optimal relational contract under $\beta=0$ and $\beta=1$.

Choice 1: $\beta=0$.

By Lemma A.1, $U^1=w-c+\delta\left[(1-\alpha_1)U^1+\alpha_1\overline{U}\right]$. By Lemma A.3, the buyer optimally chooses w subject to a binding (P-IC-e1), which gives $w_{DS}=\frac{1}{\delta(1-\alpha_1)}c+(1-\delta)\overline{U}$.

Choice 2: $\beta=1$.

By Lemmas A.2 and A.3, the buyer's optimization problem becomes $\min_w w$ subject to (P-IC-e1), (P-IC0), (1), (2), and $w\geq 0$.

By Lemma A.1, the Lagrangian is given by

$$L = w + \lambda_1(U^1 - (w - c + \delta((1-\alpha_1)\cdot U^1 + \alpha_1(U^0))))$$
$$+ \mu_1(w + \delta\overline{U} - U^1) + \mu_2(\overline{U} - U^0) + \mu_3(-w).$$

The Kuhn-Tucker conditions are given by $\frac{\partial L}{\partial w} = 1 + \mu_1(1 - \frac{\partial U^1}{\partial w}) - \mu_2\frac{\partial U^0}{\partial w} - \mu_3 \leq 0$; $\frac{\partial L}{\partial w}w = 0$; $\lambda_1 > 0$ $\mu_1, \mu_2, \mu_3 \geq 0$; $\mu_1(w + \delta\overline{U} - U^1) = 0$; $\mu_2(\overline{U} - U^0) = 0$; and $\mu_3 w = 0$.

By taking derivatives on both sides of equations (1) and (2) with respect to w, we get

$$\frac{\partial U^1}{\partial w} = 1 + \delta\Big[(1-\alpha_1)\frac{\partial U^1}{\partial w} + \alpha_1\beta\frac{\partial U^0}{\partial w}\Big], \tag{25}$$

$$\frac{\partial U^0}{\partial w} = \delta\Big[\alpha_0\frac{\partial U^1}{\partial w} + (1-\alpha_0)\beta\frac{\partial U^0}{\partial w}\Big]. \tag{26}$$

We proceed by the following steps.

Step 1: Show that $\mu_3 = 0$ and $w > 0$. Suppose that $w = 0$. Then

$$U^1 = -c + \delta\Big[(1-\alpha_1)U^1 + \alpha_1 U^0\Big] < (1-\alpha_1)U^1 + \alpha_1 U^0,$$

where the inequality comes from that $c > 0$ and $\delta < 1$. The inequality above implies that $U^1 < U^0$. Meanwhile,

$$U^0 = \delta\Big[\alpha_0 U^1 + (1-\alpha_0)U^0\Big] < \alpha_0 U^1 + (1-\alpha_0)U^0 < U^0,$$

where the first inequality comes from $\delta < 1$ and the second inequality comes from $U^1 < U^0$. Since $U^0 < U^0$ can never be true, we know that $w > 0$ and thus $\mu_3 = 0$.

The implication for $w > 0$ is that $\frac{\partial L}{\partial w} = 1 + \mu_1(1 - \frac{\partial U^1}{\partial w}) - \mu_2\frac{\partial U^0}{\partial w} = 0$.

Step 2: Discuss the values of μ_1 and μ_2.

Case 1) $\mu_1 = \mu_2 = 0$. In this case, $\frac{\partial L}{\partial w} = 0$ is violated, indicating that this case is not possible.

Case 2) $\mu_1 = 0$ and $\mu_2 > 0$. In this case, $U^1 > w + \delta\overline{U}$ and $U^0 = \overline{U}$. Solve U^1 and w based on equation (1), and we get $U^1 = \frac{1-\delta(1-\alpha_0)}{\delta\alpha_0}\overline{U}$, and $w = c + \frac{(1-\delta)^2-\delta(1-\delta)(\alpha_0+\alpha_1)}{\delta\alpha_0}\overline{U}$. Meanwhile, $\frac{\partial L}{\partial w} = 0$ and $\mu_1 = 0$ imply that $1 = \mu_2 \frac{\partial U^0}{\partial w}$.

There are two conditions that need to be satisfied for this case to be feasible and optimal. First, for feasibility, the solved w and U^1 need to satisfy $U^1 > w + \delta\overline{U}$. After plugging in U^1 and w as functions of $\overline{U}$, this requires that $\frac{(1-\delta)\overline{U}}{c} > \frac{\alpha_0}{1-\alpha_1}$. Second, since we are considering the case where choosing $\beta = 1$ is weakly better than choosing $\beta = 0$, the solved w needs to be lower than w_{DS}, which gives $\frac{(1-\delta)\overline{U}}{c} \leq \frac{\alpha_0}{1-\alpha_1}$. These two conditions contradict each other, indicating that this case is not possible.

Case 3) $\mu_1 > 0$ and $\mu_2 > 0$. In this case, both incentive compatibility constraints bind, which give $U^1 = w + \delta\overline{U}$ and $U^0 = \overline{U}$. Under binding (P-IC-e1) and (P-IC0), equations (1) and (2) can be satisfied only if $\frac{(1-\delta)\overline{U}}{c} = \frac{\alpha_0}{1-\alpha_1}$. In that case, the solved w also coincides with w_{DS}, and a buyer is thus indifferent between $\beta = 1$ and $\beta = 0$.

Case 4) $\mu_1 > 0$ and $\mu_2 = 0$. In this case, $U^1 = w + \delta\overline{U}$. Solve U^1 and w based on equation (1), and we obtain $w = w_{DI} \equiv \left(\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1)+\frac{\delta}{1-\delta}\alpha_0}\right)c + \left(\frac{1+\frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1)+\frac{\delta}{1-\delta}\alpha_0} - \delta\right)\overline{U}$.

In sum, if a buyer chooses $\beta = 1$, she pays w_{DI} when her demand is 1.

The comparison between choice 1 and choice 2 hinges on comparing w_{DS} and w_{DI}. It can be shown that $w_{DS} < w_{DI}$ if and only if $\frac{(1-\delta)\overline{U}}{c} > \frac{\alpha_0}{1-\alpha_1}$. Therefore, whenever the condition above holds, the buyer chooses $\beta = 0$ (separating from the producer when demand becomes 0) and pays w_{DS} when her demand is 1. Otherwise, she chooses $\beta = 1$ (retaining the producer when demand becomes 0) and pays w_{DI} when her demand is 1.

A.6 Proof of Optimal Managed Contract, Lemma 3

The proof proceeds by mapping the manager's problem into the direct contracting framework of Lemma 2. We verify each element of this mapping.

Step 1: The manager's incentive problem. In a managed contract, the buyer pays the manager a service fee p when $d_t = 1$. The manager's "effort" is to assign a producer and pay the competitive wage w_M, at a cost of w_M to the manager. If the manager shirks (does not assign a producer), she keeps p but does not pay w_M. Therefore, the manager's continuation payoffs take the same form as equations (1)–(2) with the substitutions $c \to w_M$ and $w^1 \to p$.

Step 2: The manager's outside option. Since matching between buyers and managers is non-anonymous, a manager who separates from a buyer cannot be re-matched with another buyer. Therefore, the manager's continuation value after separation is $\overline{V} = 0$.

Step 3: Optimal retention. Applying the criterion from Lemma 2, the buyer retains the manager ($\beta = 1$) if and only if $(1-\delta)\overline{V}/w_M \leq \alpha_0/(1-\alpha_1)$. Since $\overline{V} = 0$, the left-hand side equals zero, so this condition holds for all α_0 and α_1. Therefore, the buyer always retains the manager.

Step 4: Optimal service fee. Since $\beta = 1$ and $\overline{V} = 0$, Lemma A.2 applies (with the manager in the role of the producer), so the buyer pays the manager zero when $d_t = 0$. The service fee when $d_t = 1$ is determined by replacing c with w_M and $\overline{U}$ with 0 in the expression for w_{DI}:

$$p(\alpha_i) = \frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1) + \frac{\delta}{1-\delta}\alpha_0} w_M.$$

Step 5: Buyer's incentive constraints. The buyer's incentive constraints (B-IC-w1), (B-IC-w0), (B-IC1), and (B-IC0) take the same form as in the direct contract, with p replacing w^1 and with the buyer's continuation payoffs adjusted accordingly. Since $\overline{V} = 0$ implies the manager earns zero rents after separation, these constraints are equivalent to those verified in Lemma 2 under the substitution $c \to w_M$ and $\overline{U} \to 0$.

A.7 Proof of Optimal Contractual Choice, Proposition 1

Step 1: Compare direct contract with separation and direct contract with idleness. From Lemma 2, it is immediate that if $\overline{U} > \overline{U}^{ei} \equiv \frac{c}{1-\delta}\alpha_0$, there exists a unique cutoff $\overline{\alpha}^{1ei} \in (0,1)$ such that $w_{DS} < w_{DI}$ if and only if $\alpha_1 < \overline{\alpha}^{1ei}$.

Step 2: Compare direct contract with separation with managed contract. Observe that $w_{DS} < p$ if and only if $f(\alpha_1) \equiv \frac{1}{1-\alpha_1}\frac{c}{\delta} + (1-\delta)\overline{U} - \frac{1}{\delta - \frac{\delta}{1+\frac{\delta}{1-\delta}\alpha_0}\alpha_1}(\frac{c}{\delta} + (1-\delta)\overline{U}) < 0$.

Observe that, when $\alpha_0 = 0$, $f(\alpha_1)\big|_{\alpha_0=0} = (1-1/\delta)\frac{c}{\delta}\frac{1}{1-\alpha_1} + (1 - \frac{1}{\delta(1-\alpha_1)})(1-\delta)\overline{U} < 0$. In this case, $w_{DS} < p$ for sure, so $\overline{\alpha}^{1eo} = 1$. When $\alpha_0 > 0$, observe that $f(0) = (1-\frac{1}{\delta})\kappa < 0$, and $f(1)$ goes to infinity. We now show that the continuous function $f(\alpha_1)$ intersects with 0 only once. Note that $\frac{\partial f(\alpha_1)}{\partial \alpha_1} = \frac{\zeta(\zeta\frac{c}{\delta}-\kappa)\alpha_1^2 - 2(c-\kappa)\zeta\alpha_1 + (\delta c - \zeta\kappa)}{(1-\alpha_1)^2(\delta-\zeta\alpha_1)^2}$, where $\zeta = \frac{\delta}{1+\frac{\delta}{1-\delta}\alpha_0}$ and $\kappa = \frac{c}{\delta} + (1-\delta)\overline{U}$. Observe that the numerator is a quadratic in α_1. The coefficient of α_1^2 is $\zeta(\zeta c/\delta - \kappa) < 0$ (since $\zeta < \delta$ and $\kappa > c/\delta$), so the quadratic is concave. The denominator $(1-\alpha_1)^2(\delta-\zeta\alpha_1)^2$ is strictly positive for $\alpha_1 \in (0,1)$. Therefore, $\partial f/\partial\alpha_1$ is first increasing then decreasing (or monotonically signed). Combined with $f(0) < 0$ and $f(1) \to +\infty$, the function f crosses zero exactly once: if f is initially decreasing it must eventually increase (driven by the $(1-\alpha_1)^{-1}$ term as $\alpha_1 \to 1$), and it can cross zero only once because the concavity of the derivative's numerator permits at most one sign change in $\partial f/\partial\alpha_1$. Let the unique zero-crossing be $\overline{\alpha}^{1eo} \in (0,1)$.

In sum, there exists a unique cutoff $\overline{\alpha}^{1eo} \in (0,1]$ such that $w_{DS} < p$ if and only if $\alpha_1 < \overline{\alpha}^{eo}$.

Step 3: Compare direct contract with idleness and managed contract. Observe that $w_{DI} < p$ if and only if $g(\alpha_1) \equiv \frac{1+\frac{\delta}{1-\delta}\alpha_0}{1+\frac{\delta}{1-\delta}\alpha_0-\alpha_1}(\frac{2\delta-1}{\delta}\overline{U} - \frac{1-\delta}{\delta}\frac{c}{\delta}) < \delta\overline{U}$. Observe that $g(0) = \frac{2\delta-1}{\delta}\overline{U} - \frac{1-\delta}{\delta}\frac{c}{\delta}$ and $g(1) = \frac{1+\frac{\delta}{1-\delta}\alpha_0}{\frac{\delta}{1-\delta}\alpha_0}(\frac{2\delta-1}{\delta}\overline{U} - \frac{1-\delta}{\delta}\frac{c}{\delta})$. Also observe that $g(0) - \delta\overline{U} = \frac{-(1-\delta)^2}{\delta}\overline{U} - \frac{1-\delta}{\delta}\frac{c}{\delta} < 0$. If $\overline{U} < \overline{U}^{io} \equiv \frac{(1-\delta(1-\alpha_0))c}{\delta(\delta(2-(1-\delta)\alpha_0)-1)}$, $g(1) < \delta\overline{U}$ and thus the inequality holds for sure. Otherwise, since $g(\alpha_1)$ is strictly increasing in α_1, by the intermediate value theorem, there exists a unique cutoff $\overline{\alpha}^{1io} \in (0,1)$ such that $w_{DI} < p$ if and only if $\alpha_1 < \overline{\alpha}^{1io}$. Otherwise, $w_{DI} < p$ for sure.

Step 4: Put it together. Let $\overline{U}_* = \overline{U}^{io}$ and $\alpha_* = \min\{\overline{\alpha}^{1io}, \overline{\alpha}^{1eo}\}$. If $\overline{U} < \overline{U}^{io}$, $w_{DI} < p$ by Step 3, so the managed contract is dominated by the direct contract with idleness whenever the latter is feasible. Moreover, since $w_{DS} < p$ by Step 2 for $\alpha_1 < \overline{\alpha}^{1eo}$, the direct contract with separation also dominates when applicable. Hence, no buyer chooses the managed contract.

If $\overline{U} \geq \overline{U}^{io}$, the contract regions are determined as follows. By Step 1, the boundary between direct-with-separation and direct-with-idleness is $\overline{\alpha}^{1ei}$. By Steps 2 and 3, the managed contract dominates the relevant direct contract for $\alpha_1 > \alpha_*$. We verify consistency: at $\overline{U} = \overline{U}^{io}$, Steps 1–3 imply $\overline{\alpha}^{1ei} = \overline{\alpha}^{1eo} = \overline{\alpha}^{1io}$, so the three boundaries coincide. For $\overline{U} > \overline{U}^{io}$, we have two cases. If $w_{DS} < w_{DI}$ (which occurs when $\alpha_1 < \overline{\alpha}^{1ei}$), the managed contract is optimal when $p < w_{DS}$, namely when $\alpha_1 > \overline{\alpha}^{1eo}$. If $w_{DS} > w_{DI}$ (which occurs when $\alpha_1 > \overline{\alpha}^{1ei}$), the managed contract is optimal when $p < w_{DI}$, namely when $\alpha_1 > \overline{\alpha}^{1io}$.

A.8 Proof of Cross-sectional Comparisons, Corollary 1

First, note that $w_D(\alpha_i) > w_M$ for all $i \in \mathscr{I}_{DS} \cup \mathscr{I}_{DI}$. Therefore, $E[w_D(\alpha_i) \mid i \in \mathscr{I}_{DS} \cup \mathscr{I}_{DI}] > E[w_M | i \in \mathscr{I}_M]$. Second, note that $w_D(\alpha_i)$ takes on different values depending on α_i, which is heterogeneous across buyers, while w_M is constant. Therefore, $\mathrm{Var}[w_D(\alpha_i) \mid i \in \mathscr{I}_{DS} \cup \mathscr{I}_{DI}] > \mathrm{Var}[w_M \mid i \in \mathscr{I}_M] = 0$. Third, the separation rate for buyer i is $(1-\beta_i)\alpha_{1i}$: a match ends when demand switches from 1 to 0 (which occurs with probability α_{1i}) and the buyer separates (which occurs when $\beta_i = 0$). Note that $\beta_i = 0$ for all $i \in \mathscr{I}_{DS}$, so these buyers have separation rate $\alpha_{1i} > 0$. Meanwhile, $\beta_i = 1$ for all $i \in \mathscr{I}_{DI} \cup \mathscr{I}_M$, so these buyers have separation rate 0. It follows that directly contracted producers (weakly) have higher separation rates than managed producers. Fourth, the idleness for a buyer $i \in \mathscr{I}_{DI}$ is given by $(1-\pi_i) > 0$, where $\pi_i = \alpha_{0i}/(\alpha_{0i} + \alpha_{1i})$ is the steady-state probability of positive demand. By contrast, for $i \in \mathscr{I}_{DS} \cup \mathscr{I}_M$, idleness is zero: under $\mathscr{I}_{DS}$, the buyer separates when demand is zero; under $\mathscr{I}_M$, the manager (not the producer) bears the idle state.

A.9 Proof of Managerial Impacts, Corollary 2

Since $\overline{U} = C$ by the free-entry condition, the introduction of managers does not change $\overline{U}$. Therefore, the payment $w_D(\alpha_i)$ for buyers who continue to use direct contracts is unchanged.

With the introduction of managers, the per-period payoff of buyer $i \in \mathscr{S}^0$ increases by $y - p(\alpha_i)$. The per-period payoff of buyer $i \in \mathscr{S}_I$ increases by $w_{DI}(\alpha_i) - p(\alpha_i)$. The per-period payoff of buyer $i \in \mathscr{S}_S$ increases by $w_{DS}(\alpha_i) - p(\alpha_i)$. The per-period payoff of the remaining buyers are unchanged. The average producer pay per unit effort falls, since $w_D(\alpha_i) > w_M$ for all $i \in \mathscr{S}_I \cup \mathscr{S}_S$, and these buyers switch from paying $w_D(\alpha_i)$ per unit of service to paying w_M through the manager. Producer separation rates and idleness also fall, since buyers switch from direct contracts with either idleness or separation to managed contracts in which buyers are never idle and never separate. The measure of buyers who receive services increases by $\left|\mathscr{S}^0\right|$. The total measure of services provided in the economy increases by $\int_{\mathscr{S}^0} \pi_i dF$. However, for $i \in \mathscr{S}_I$, the steady-state measure of producers matched to these buyers fall, since idleness falls. The total measure of matched producer $(n_M + n_B)$ therefore increases if and only if $\int_{\mathscr{S}^0} \pi_i dF > \int_{\mathscr{S}_I} (1-\pi_i) dF$.

B Model with Producer Specialization

B.1 Proof of Equilibrium with Specialization, Proposition 3

In this extension, demand switching is symmetric: $\alpha_{0i} = \alpha_{1i} \equiv \alpha_i$. Thus α_i plays the role of both α_{0i} and α_{1i} from the main model.

Based on Lemma 2 and 3, the buyer's post-matching continuation payoffs when she has demand is $\Pi_D(\alpha_i) = \frac{y+\Delta_i}{1-\delta} - \left[\overline{U} + \frac{c}{(1-\alpha_i)\delta(1-\delta)}\right]$ if directly contracting with a specialist producer. It is $\Pi_G = \frac{y}{1-\delta} - \left[\overline{U} + \frac{c}{\delta(1-\delta)}\right]$, if directly contracting with a generalist producer. It is $\Pi_M(\alpha_i) = \frac{y+\Delta_i}{1-\delta} - \left[\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i - \alpha_i}(\overline{U} + \frac{c}{\delta(1-\delta)})\right]$ if contracting with a manager.

Three observations follow. First, a buyer prefers to directly contract with a specialist than directly contract with a generalist if and only if $\Pi_D(\alpha_i) \geq \Pi_G$, or, $\Delta_i \geq \frac{\alpha_i}{1-\alpha_i}\frac{c}{\delta}$. Second, a buyer prefers to directly contract with specialists rather than contract with a manager if and only if $\left[\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i - \alpha_i} - 1\right]\left[\overline{U} + \frac{c}{\delta(1-\delta)}\right] \geq \frac{\alpha_i}{1-\alpha_i}\frac{c}{\delta(1-\delta)}$. By Proposition 1, there exists a unique cutoff α^{ei} such that the buyer prefers to contract with a manager if and only if $\alpha_i > \alpha^{ei}$. Third, a buyer prefers a managed contract over direct contract with a generalist if and only if $\Delta_i \geq \overline{\Delta}(\alpha_i) \equiv \left[\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i - \alpha_i} - 1\right]\left[\overline{U} + \frac{c}{\delta(1-\delta)}\right]$. Therefore, the buyer choose to contract with a manager if and only if α_i and Δ_i are both large enough.

B.2 Proof of Managerial Impacts with Specialization, Corollary 3

Let $\mathscr{S}_G$ denote buyers who switch from direct contracts with generalists to managed contracts with specialists when managers become available. Let $\mathscr{S}_B$ be the set of buyers who switch from direct contracts with specialists to managed contracts with specialists. Given the assumption on the distribution of (Δ_i, α_i), $|\mathscr{S}_G|, |\mathscr{S}_B| > 0$. The contractual choice of the remaining buyers are unchanged. Therefore, the measure of specialists unambiguously increase with managerial coordination.

The Bellman equation for unmatched producers can be rewritten as $\overline{U} = \frac{v}{u}\frac{E[v_i U_{1i}]}{v} + \left(1 - \frac{v}{u}\right)\delta\overline{U}$. With the introduction of managers, more producers are contracted as specialists under managed contracts without separation, so v (the total measure of vacancies arising from separations) falls. Moreover, the average continuation value offered by vacancies, $E[v_i U_{1i}]/v$, also falls because managed contracts pay $w_M < w_D(\alpha_i)$, so U_{1i} is lower for managed producers. Since $\overline{U} = C$ is pinned down by free entry, the Bellman equation requires v/u to increase to compensate for the lower per-vacancy value. Given that v falls and v/u rises, it follows that u falls by a larger proportion than v. Since the total producer population adjusts through entry and $n_N = u - v$ represents unmatched producers net of vacancies, n_N falls: fewer producers are unmatched in equilibrium.

C Model with Exogenous Producer Population

The main text assumed that the number of producers is determined via endogenous entry. In this appendix, we analyze an economy where the number of producers is fixed exogenously at $n > 1$. The main difference is that when the producer population is fixed, the presence of managers alters $\overline{U}$ by changing the overall demand for producers, and thereby alters equilibrium outcomes.

We first show that if n is sufficiently large, then the producer market becomes very slack, buyers prefer to enter contracts with idleness, and there is no managerial coordination.

Proposition C.1. *Assume the number of producers n is exogenously fixed. There exists* *a unique steady-state equilibrium. If n is sufficiently large, then there are no managed contracts.*

Proof. Given $\overline{U}$, we can compare the values of $w_{DS}(\alpha_i)$, $w_{DI}(\alpha_i)$, $p(\alpha_i)$, and y for each i using (6), (7), and (11) to determine $\mathscr{I}_{DS}$, $\mathscr{I}_{DI}$, and $\mathscr{I}_M$. Having derived these, we can obtain unique values for v_i, v, n_B, and n_M from (12), (13), (14), and (15). Plugging $w_{DS}(\alpha_i)$, $w_{DI}(\alpha_i)$, and $w_M(\alpha_i)$ into (18) defines a mapping $T(\overline{U})$. We claim that T has a unique fixed point. First, T is continuous: although the contract choice sets $\mathscr{I}_{DS}$, $\mathscr{I}_{DI}$, $\mathscr{I}_M$ change discontinuously at the threshold values of $\overline{U}$, the buyer's payoff is continuous at each threshold (since the buyer is indifferent at the switching point). Second, $T(0) > 0$ since producers matched to buyers with positive demand earn positive rents. Third, $T(\overline{U}) < \overline{U}$ for $\overline{U}$ sufficiently large, since all terms entering the Bellman equation are discounted by $\delta < 1$ and wages are bounded above by y. By the intermediate value theorem, a fixed point exists. Uniqueness follows because T is weakly increasing in $\overline{U}$ with slope strictly less than one, since the Bellman equation discounts all future payoffs by $\delta < 1$. By (16), (17), and (18), $\overline{U}$ is decreasing in n, and $\overline{U} \to 0$ as $n \to \infty$. The desired result follows by Proposition 1. □

We next compare economies with and without managers assuming that n is fixed and that some nonzero measure of buyers choose managerial coordination when possible. In this economy, introducing managers has both direct and indirect effects. The direct effect holds $\overline{U}$ fixed and was characterized in the previous subsection. Due to this effect, various types of buyers switch to managerial coordination, producer rents fall, and the number of matched producers may or may not increase. Without free entry, however, $\overline{U}$ also adjusts. A change in $\overline{U}$ affects all of the endogenous variables in the model, so managerial coordination has indirect spillover effects onto all producers.

The sign of the resulting change in $\overline{U}$ depends on the distribution of buyer types α_i. If a large set of buyers switch from being unmatched to matched, then $\overline{U}$ would increase due to reduced wait time in a tightened producer market. This indirect effect raises the pay of all producers. On the other hand, if no buyers switch from being unmatched to being matched, then $\overline{U}$ would unambiguously fall due to producer rent reductions associated with initially matched buyers switching to managerial coordination. This indirect effect instead reduces the pay of all producers.